# The Cheon–Hwang Sub-Dittert Conjecture at $k = 3$ and $k = 4$

with a stability form of the Tverberg–Friedland theorem
and a decomposition of the deficit uniform in $n$ and $k$

D C P Revere

dcprevere@gmail.com

*Draft — 30 July 2026*

**Abstract.** Cheon and Hwang conjectured in 1992 that $E_k(r) + E_k(c) - P_k(A) \le 2 - k!/n^k$ for every non-negative $n \times n$ matrix of total sum $n$ and every $1 \le k \le n$, with equality only at $J_n/n$; the endpoint $k = n$ is Dittert's conjecture. This paper works the intermediate range $2 < k < n$ on two lines, and proves a stability theorem that the second line consumes. The results rest on different support layers — kernel-checked in Lean 4, or written proofs with an exact rational verifier — and each claim below is graded explicitly; the grading is part of the claim.

**The line $k = 3$, every $n \ge 4$**: the inequality, the equality case, and the quantitative strengthening

$$(2 - 6/n^3) - [E_3(r) + E_3(c) - P_3(A)] \ge \theta_2(n) \cdot \|A - J_n/n\|_F^2, \quad \theta_2(n) = \frac{n^4 + 40n^2 - 84n + 40}{n^5(n-1)^3(n-2)},$$

by a single Positivstellensatz certificate whose nineteen symmetry-reduced coefficients are explicit rational functions of $n$: the governing degree is $k$, not $n$, so the reduced program has the same $12 \times 19$ shape at every dimension and is solved once over $\mathbb{Q}(n)$, with definiteness of the two $n^2 \times n^2$ Grams reduced in closed form to ten rational functions of $n$ decided by Sturm sequences. This part is machine-checked end to end (`subDittert_k3_full`): no `sorry`, Lean's standard axioms. With Hwang's 1987 theorem at $n = 3$, the line $k = 3$ is settled in every dimension.

**A stability form of the Tverberg–Friedland theorem.** On the doubly stochastic polytope the excess of $\sigma_k$ over its minimum at $J_n/n$ is at least $\binom{n}{k}^2 c(n,k) \|A - J_n/n\|_F^2$ with $c(n,k) = k(k-1)\,k!/\big(4n^k(n-1)^2\big)$, for $k = 2$ (every $n \ge 2$), $k = 3$ (every $n \ge 4$), $k = 4$ (every $n \ge 8$) and $k = 5$ (every $n \ge 14$); the constant is optimal to within a factor two, and the cell $(k,n) = (3,3)$ is a genuine exception. Kernel-checked at $k = 2$, $k = 3$ and $k = 4$, in each case over the whole range the statement covers for that $k$; written proofs plus an exact verifier at $k = 5$.

**The line $k = 4$**: the Cheon–Hwang inequality holds for every $n \ge 10$, with equality only at $J_n/n$, by a local route — confinement traps any violator in a collar of the Birkhoff polytope, a collar matrix splits orthogonally into a line-sum block and a doubly centred block, and the five cross terms between the blocks have exact reductions. A companion sensitivity computation, graded as exactly that, finds the threshold unmoved at every sampled admissible value of the one constant in the argument that is not settled, with the shape of the $c$-dependence asserted mechanically. The theorem is not formalised; its support is written proofs plus an exact verifier that re-reads the displayed numbers out of this document. Fixed-dimension certificates settle the cells $(5,4)$, $(6,4)$ and $(7,4)$ besides, so on the line $k = 4$ exactly two cells remain open — $n = 8$ and $n = 9$ — and each is recorded at the grade it has actually reached.

These results sit inside an exact decomposition of the deficit uniform in $n$ and $k$, from which the classical case $k = 2$ follows in a few lines with a machine-checked proof. We also formalise, for every $2 \le k \le n$, the confinement of any violator to a shrinking neighbourhood of the Birkhoff polytope, together with the Newton and Maclaurin inequalities that the confinement consumes, neither of which is in Mathlib. Kernel-checked results are cited by Lean declaration name at a pinned commit.

# 1 Introduction

Let

$$K_n = \left\{ A \in \mathbb{R}^{n\times n} : A_{ij} \geq 0, \sum_{i,j} A_{ij} = n \right\}, \tag{1}$$

and for $A \in K_n$ let $r$ and $c$ be its vectors of row and column sums. Write $\sigma_k(A)$ for the sum of the permanents of all $k \times k$ submatrices of $A$ — rows chosen from one $k$-subset, columns from another, independently — and put

$$E_k(v) = \frac{e_k(v)}{\binom{n}{k}}, \quad P_k(A) = \frac{\sigma_k(A)}{\binom{n}{k}^2}, \quad \gamma(n,k) = \frac{k!}{n^k}, \tag{2}$$

with $e_k$ the elementary symmetric function. Write $\Phi_k(A) = E_k(r) + E_k(c) - P_k(A)$.

**The conjecture (Cheon and Hwang [1], 1992).** For every $A \in K_n$ and every $1 \leq k \leq n$,

$$\Phi_k(A) \leq 2 - \gamma(n,k), \tag{3}$$

with equality only at $A = J_n/n$.

The endpoints are familiar. At $k = n$ one has $\sigma_n = \mathrm{per}$ and $E_n(v) = \prod_i v_i$, so the statement is $\prod_i r_i + \prod_j c_j - \mathrm{per}(A) \leq 2 - n!/n^n$: this is **Dittert's conjecture** verbatim, Conjecture 28 of the Cheon–Wanless survey [5]. At $k = 1$ the inequality is an identity **on $\boldsymbol{K_n}$** — not on all of $\mathbb{R}^{n\times n}$ — since the difference is a constant multiple of $\sum_{ij} A_{ij} - n$; we note the distinction because a test that treats it as a formal identity fails. The cases $k \leq 2$ for every $n$, and every $k$ at $n \leq 3$, are known.

The interesting range is therefore $2 < k < n$, and it is that range which appears to have gone untouched. This paper settles the whole line $k = 3$.

## 1.1 Results

**Theorem A ($\boldsymbol{k = 3}$, every dimension).** For every $n \geq 4$ and every $A \in K_n$,

$$E_3(r) + E_3(c) - P_3(A) \leq 2 - \frac{6}{n^3}, \tag{4}$$

with equality if and only if $A = J_n/n$. More precisely, writing $\|\cdot\|_F$ for the Frobenius norm,

$$\begin{gathered}\left(2 - \frac{6}{n^3}\right) - [E_3(r) + E_3(c) - P_3(A)] \geq \theta_2(n) \cdot \|A - J_n/n\|_F^2, \\ \text{where} \quad \theta_2(n) = \frac{n^4 + 40n^2 - 84n + 40}{n^5(n-1)^3(n-2)},\end{gathered} \tag{5}$$

and $\theta_2(n) > 0$ for every $n \geq 4$.

Lean: `SubDittertK3.subDittert_k3_full`.

The three statements are one statement. Equation 5 implies Equation 4 because $\theta_2(n) > 0$, and it implies the equality case because $\|A - J_n/n\|_F = 0$ only at $J_n/n$; conversely $\Phi_3(J_n/n) = 2 - 6/n^3$ exactly. We state all three because the quantitative form is what the certificate actually produces, and it is stronger than what the conjecture asks for: the functional falls away from its maximum at least quadratically in the distance to the maximiser, at an explicit rate. The constant $\theta_2(n)$ is not claimed optimal; see §9.4.

**Corollary B (two layers, stated as such).** With Hwang's theorem for Dittert at $n = 3$ [3], the case $k = 3$ of the Cheon–Hwang conjecture holds, with its equality case, for every $n \geq 3$. Since $k = 3$ requires $n \geq 3$, no dimension is left open on this line.

**This corollary is not a single-layer result and should not be cited as one.** The range $n \geq 4$ is Theorem A and is machine-checked (`SubDittertK3.subDittert_k3_full`). The remaining case $n = 3$ is Hwang's refereed theorem of 1987 [3], which we do not reprove in Lean. The corollary's support is therefore Lean **plus** the published literature, and it inherits whatever reliance one places on [3].

Since $2 < 3 < n$ for every $n \geq 4$, Theorem A lies inside the intermediate range. On the evidence of §2 it is the first resolved case of the Cheon–Hwang conjecture with $2 < k < n$. That is a priority claim rather than a claim of difficulty, and §2 records why it is perishable.

The certificate is a construction special to $k = 3$, but it sits inside a decomposition that is not.

**Theorem C (the deficit, decomposed uniformly in $n$ and $k$).** Write $A = J_n/n + b$, let $R$ and $C$ be the row and column sums of $b$, and put

$$s_d = \frac{[k]_d}{[n]_d}, \quad t_d = s_d^2 \cdot \frac{(k-d)!}{n^{k-d}} \tag{6}$$

with $[x]_d$ the falling factorial. Then for every $1 \leq k \leq n$, identically in $b$,

$$(2 - \gamma(n,k)) - \Phi_k(A) = \sum_{d=1}^{k} [t_d \cdot \sigma_d(b) - s_d \cdot (e_d(R) + e_d(C))]. \tag{7}$$

Lean: `SubDittertUniversal.universal_identity`.

Equation 7 is elementary, and no novelty is claimed for it: it is the $\sigma_k$ analogue of the classical expansion $\operatorname{per}(A + xJ_n) = \sum_j x^{n-j}(n-j)!\,\sigma_j(A)$, combined with $e_k(\mathbf{1} + R) = \sum_d \binom{n-d}{k-d} e_d(R)$. Its role here is that it makes every $k$-dependence explicit and finite, so that statements which look like separate facts at separate $k$ become instances of one identity. The absence of a $d = 0$ term is the reason the constant $2 - \gamma(n,k)$ is exactly right, at every $n$ and $k$ at once.

**Theorem D ($k = 2$, every dimension).** For every $n \geq 2$ and every $A \in K_n$,

$$\Phi_2(A) \leq 2 - 2/n^2. \tag{8}$$

With $\kappa = 2/(n(n-1))$ the deficit is a manifest sum of squares on the hyperplane $\sum_{ij} b_{ij} = 0$:

$$(2 - 2/n^2) - \Phi_2(A) = \frac{1}{2}\left[\kappa^2 \|b\|^2 + \kappa(1-\kappa)(\|R\|^2 + \|C\|^2)\right]. \tag{9}$$

Lean: `SubDittertK2.subDittert_k2`. **The statement is classical** (§2.2); the contributions are the derivation from Equation 7 and the machine-checked proof.

Theorem D needs no Cauchy–Schwarz, no Gram matrix and no Maclaurin inequality: Equation 9 falls out of Equation 7 at $k = 2$, and both coefficients are non-negative for $n \geq 2$ because $\kappa \leq 1$ there. Its interest is as a control on the uniform machinery — it is the one case where that machinery can be run end to end against an answer that is independently known — and, with Theorem A, it makes the band $k \in \{2, 3\}$ complete.

Two further results are recalled rather than claimed. Both are formalised here, and it is the formalisation, not the mathematics, that is ours.

**Theorem E (confinement; Cheon and Wanless [6], Theorem 2.1, at $k = n$; transferred to all $k$).** Let $2 \le k \le n$ and $A \in K_n$. Then

$$\frac{\|r - \mathbf{1}\|^2 + \|c - \mathbf{1}\|^2}{n(n-1)} - \gamma(n,k) \le (2 - \gamma(n,k)) - \Phi_k(A), \tag{10}$$

so any $A$ violating the Cheon–Hwang bound satisfies

$$\|r - \mathbf{1}\|^2 + \|c - \mathbf{1}\|^2 \le \frac{(n-1)k!}{n^{k-1}}. \tag{11}$$

Lean: `SubDittertMaclaurin.theoremM'`, `SubDittertMaclaurin.confinement'`.

At $k = n$ this is Theorem 2.1 of [6] in contrapositive form, with subset sums in place of the $\ell^2$ norm and a threshold of the same scale; that paper describes its own result as significantly extending a result of Hwang, so the idea is older still. Pang [9] runs the same argument at $k = n$ and refines the quantitative step without changing its shape; that item is an unrefereed preprint. The only part not found in the literature we checked is the transfer from the Dittert functional to the whole Cheon–Hwang family uniformly in $k$, **and that is a routine transfer of a known method**. It is formalised here for two reasons that have nothing to do with novelty: it is elementary enough to check, and it is uniform in $k$, so one Lean proof covers the whole family.

**Theorem F (Newton; Maclaurin).** For every real-rooted real polynomial and every admissible index, Newton's inequality holds; in vector form, for $v \in \mathbb{R}^n$ and $1 \le j \le n-1$,

$$e_{j-1}(v)\, e_{j+1}(v) \binom{n}{j}^2 \le e_j(v)^2 \binom{n}{j-1}\binom{n}{j+1}, \tag{12}$$

with no positivity hypothesis. Consequently, for $v \ge 0$ with $\sum_i v_i = n$ and $2 \le k \le n$, $E_k(v) \le E_2(v)$.

Lean: `NewtonIneq.newtonAt_all`, `NewtonIneq.newton_esymF`, `NewtonIneq.pnorm_le_two`. **Classical mathematics**; the contribution is the formalisation.

The telescoped form $E_k \le E_2$ carries no fractional powers, unlike the usual $E_k^{1/k} \le E_2^{1/2}$; that is what makes it the statement to formalise, and it is exactly what Theorem E consumes. Neither Newton's inequalities nor Maclaurin's inequality is in Mathlib v4.14.0, whose `NewtonIdentities` file carries Newton's **identities** only.

The remaining results move off the line $k = 3$. The first lives on the doubly stochastic polytope and is a quantitative form of a theorem conjectured by Tverberg [17] and proved by Friedland [18]: $\sigma_k$ on $\Omega_n$ attains its minimum only at the barycentre. The second consumes it.

**Theorem G (stability of the Tverberg–Friedland minimum).** Let $2 \le k \le 5$ and let $n$ satisfy

$$n \ge 2 \text{ at } k = 2, \quad n \ge 4 \text{ at } k = 3, \quad n \ge 8 \text{ at } k = 4, \quad n \ge 14 \text{ at } k = 5. \tag{13}$$

Then for every doubly stochastic $n \times n$ matrix $A$,

$$\sigma_k(A) - \binom{n}{k}^2 \frac{k!}{n^k} \geq \binom{n}{k}^2 c(n,k)\, \|A - J_n/n\|_F^2, \quad c(n,k) = \frac{k(k-1)\,k!}{4n^k(n-1)^2}. \tag{14}$$

The constant is optimal to within a factor two (§10.8), and the cell $(k,n) = (3,3)$ is a genuine exception with an explicit witness (§10.7).

Support, split by cell. **Kernel-checked** at ($k = 2$, every $n \geq 2$) and ($k = 3$, every $n \geq 4$) — Lean: `stabilityAt_two` and `stabilityAt_three` in `StabilityK3.lean` at commit `1507013` — and at ($k = 4$, every $n \geq 8$) — Lean: `stabilityAt_four` in `StabilityK4.lean` at commit `944b517` — in each case the whole range Theorem G states for that $k$, with no arithmetic gap. **Written proofs plus the exact verifier of §10.9** at ($k = 5, n \geq 14$): no `stabilityAt_five` exists. §10.10 states the division precisely, with the checks that keep it honest, and states two limits of the $k = 4$ formalisation that must not be over-read.

**Theorem H ($k = 4$, every $n \geq 10$).** For $k = 4$ and every $n \geq 10$, $\Phi_4(A) \leq 2 - 24/n^4$ for every $A \in K_n$, with equality only at $A = J_n/n$.

Support: written proofs (§11) plus the exact verifier `graded_verify_k4.py`, which recomputes every displayed quantity of that part over $\mathbb{Q}$ with no floating-point arithmetic in any decision, and parses the sensitivity table of §11.8 out of this document so that displayed and checked cannot drift apart. **Not formalised**; §11.1 states the scope exactly.

**Insensitivity of the threshold (a verified sensitivity computation, not a theorem).** Write $q_i(z) \leq c\,(1 - 1/n)$ for the collar bound on the row squared norms of the doubly centred block, $c \geq 1$ the one unsettled constant of the $k = 4$ argument. At each of the ten values $c \in \{1.00, 1.25, 1.50, 1.58, 1.63, 2.00, 2.34, 2.53, 3.00, 4.00\}$ the honest threshold is recomputed exactly over $\mathbb{Q}$, and it equals 10 at every sampled value inside the admissible band $1.58 \leq c \leq 2.53$, whose ends are the smallest constructed violation and the proved collar cap. A mechanical audit — every budget line recomputed at $c$ and $2c$, at $n = 10$ and $n = 16$, the run aborting on any change — asserts that exactly three budget lines depend on $c$, one linearly and two through $\sqrt{c}$, and that no other line moves.

Support: the named $c$-set plus that structural audit, both under the verifier of §11.9; no claim is made for unsampled $c$, and the grade sentence is the claim. This computation is why we regard Theorem H's threshold as stable: no sampled admissible value moves it, and the audited structure names the only places $c$ can act.

Between Theorem A's line and Theorem H's range sit the five cells ($k = 4, 5 \leq n \leq 9$), covered by neither theorem. Three of them are settled by fixed-dimension certificates at full anchor grade — $(5,4)$, $(6,4)$ with $\Phi_4 \leq 107/54$ on $K_6$, and $(7,4)$ with $\Phi_4 \leq 4778/2401$ on $K_7$ — so the open cells on the line $k = 4$ are exactly $n = 8$ and $n = 9$. §12 records all five cell by cell, each at the grade it has actually reached; §11.1 states the honest gap rather than leaving it to be inferred.

## 1.2 Support layers

Every result above is stated with the layer it rests on, because in this corner of the literature the distinction has recently been expensive.

- **Theorems A and C–F are proved in Lean 4**, on Lean's standard axioms, with no `sorry` anywhere in the development and no use of `native_decide`. §9 states the verification standard precisely and pins it to a commit. Theorem A is Lean-proved in all three parts, equality case and stability bound included.

- **Theorem G splits by cell**: kernel-checked at $k = 2, 3$ (`StabilityK3.lean`, commit `1507013`) and at $k = 4$ (`StabilityK4.lean`, commit `944b517`), in each case over the whole stated range; written proofs plus the exact verifier `graded_verify_stability.py` at $k = 5$. §10.10 is the precise statement, and the verifier checks that statement itself against the Lean sources at the pinned commits.
- **Theorem H is not formalised.** Its support is written proofs plus the exact verifier `graded_verify_k4.py`; every quantity displayed in §11 is recomputed over $\mathbb{Q}$, and the sensitivity table is parsed out of this document rather than restated. The insensitivity claim beside it is a verified sensitivity computation, graded in its own statement.
- **Corollary B** rests on Theorem A for $n \geq 4$ and on the published literature at $n = 3$.
- **The fixed-dimension certificates of §7.1 are not Lean-checked.** They are exact rational data accepted by an independent standalone verifier. They are recorded as anchors; no theorem above depends on them, and the cell $(5, 4)$ among them is one of the per-cell records of §12.
- **§13 is computational evidence, not a theorem of this paper**, and is not Lean-checked. Its verdicts are exact rational computations with stored witnesses.
- **Nothing here is refereed**, including this paper. Exact verification and machine checking are different things from refereeing, and weaker ones.

# 2 The state of the line, and of the neighbouring endpoint

## 2.1 The $k = n$ endpoint moved in one week, publicly and unrefereed

Dittert's conjecture is the endpoint of the same family, and its status changed sharply in July 2026. Only two dimensions rest on refereed work: $n = 2$ (Sinkhorn [4]) and $n = 3$ (Hwang [3]). Everything else that is currently claimed is either an unrefereed preprint or a public code repository:

| dimensions | claimed by | status |
|---|---|---|
| $n = 2$ | Sinkhorn [4], 1984 | refereed |
| $n = 3$ | Hwang [3], 1987 | refereed |
| $n = 4$ | Divya K. U. and Somasundaram, arXiv:2312.00464 | **claim withdrawn in the published version** [8] |
| $n = 4$ | public repository [11], 21 Jul 2026 | public, dated, unrefereed |
| $n = 4, 5, 8–16$ | public repository [11], 23 Jul 2026 | public, dated, unrefereed |
| $n = 6–15$ | public repositories [11], 24–25 Jul 2026 | public, dated, unrefereed |
| $n = 16$ | Kafidov [10], arXiv:2607.19439, 21 Jul 2026 | preprint, no DOI |
| $n \geq 17$ | Pang [9], arXiv:2606.01531, 1 Jun 2026 | preprint, no DOI |

Table 1: Public dated claims on Dittert's conjecture ($k = n$), as of 28 July 2026. Refereed status is stated as it is, not as one would like it.

Two features of the table bear on how the present paper should be read. At the 28 July 2026 sweep none of the repositories listed [11] had a corresponding preprint or journal article, so the indexed literature does not on its own record the state of this conjecture. And the record moves in both directions: the v1 abstract of arXiv:2312.00464 claims Dittert at $n = 4$, while the published version in *Special Matrices* **12** (2024) [8] contains no such claim; the preprint record still shows v1.

None of the unrefereed items is asserted here to be wrong. They are public and dated, which is what matters for priority, and none of them is refereed, which is what matters for reliance.

One of them warrants a sentence of positioning. Taken together, the repository claims of [11] and the two preprints [9, 10] assemble a claim on Dittert's conjecture **itself**, in full — the `pedromnasc` repository presents exactly such an assembly — and no part of that assembly has been refereed, nor has the assembly as a whole. This paper takes no view on its correctness. The contrast in scope discipline is deliberate: every priority claim made here is narrowed to what the Lean development and the exact verifiers actually carry, at the grade each part actually has, and the grades are stated claim by claim rather than inherited from the strongest part.

### 2.2 The intermediate range $2 < k < n$

The generalisation itself has attracted almost nothing. The Cheon–Hwang paper [1] has five citing papers — Cheon and Wanless 2012 [6], Cheon and Wanless 2007 [7], the Cheon–Wanless survey [5], and Cheon and Yoon 2006 and Cheon 1993 (both [12]) — and none of them works on the generalisation. Reference [6] says only that Cheon and Hwang "posed a problem generalising the Dittert conjecture"; its own results concern partly decomposable matrices and asymptotics. To our knowledge neither the literature nor any public code repository carries a result on the intermediate range; the closest item is the one described next.

The one adjacent effort is a `subdittert/` package inside one of the repositories of [11]. Its README is explicit about scope, and we quote it verbatim:

> "The endpoint `k=n` is Dittert's problem. The cases `k<=2` are historically known. This package does **not** claim the unresolved intermediate cases."

That quotation is also the clearest available statement of the prior status of Theorem D: $k \le 2$ is historically known, for every $n$, and we claim no novelty for the statement. Its results are $k = n - 1$ for $18 \le n \le 80$, $k = n - 2$ for $18 \le n \le 40$, and a table over 425 pairs with $17 \le n \le 40$ — and all three are conditional on the package's own "positive Li-scaling structural hypothesis": its README states the $k = n - 1$ Hall-cut exclusion "with" that hypothesis, the $k = n - 2$ exclusion "subject to the same structural hypothesis", and the 425-pair table conditional on both it and the package's other intermediate cases. What is unconditional there is a Sturm-certified floor on the one-zero Birkhoff face at each order. Note that $(n, k) = (4, 3)$ **is** a $k = n - 1$ case, so that package works the same diagonal — but from $n = 18$ upward, by a large-$n$ scaling argument that cannot reach $n = 4$, and conditionally throughout. It does not intersect the line $k = 3$ except at $n = 4$, which it excludes.

One item is flagged rather than paraphrased: Cheon and Wanless 2007 [7] is paywalled and was not obtained, and is reported to settle $n \le 3$ for all $k$. Nothing in this paper depends on it: Corollary B takes $n = 3$ from Hwang [3], and §7.1 reproves it by certificate.

## 3 The decomposition, and the case $k = 2$

Theorem C is proved by reading the two definitions and dividing. Since $A = J_n/n + b$ has row sums $\mathbf{1} + R$, $e_k(\mathbf{1} + R) = \sum_d \binom{n-d}{k-d} e_d(R)$; and since a term of a permanent is a bijection, $\sigma_k(J_n/n + b) = \sum_d \binom{n-d}{k-d}^2 (k-d)!\, n^{-(k-d)} \sigma_d(b)$. Dividing by $\binom{n}{k}$ and $\binom{n}{k}^2$ turns $\binom{n-d}{k-d}/\binom{n}{k}$ into $[k]_d/[n]_d = s_d$. The $d = 0$ terms are 1, 1 and $\gamma(n, k)$, which cancel the constant $2 - \gamma(n, k)$ exactly; that cancellation is why the sum in Equation 7 starts at $d = 1$.

Three consequences are worth naming, because each has been observed separately in this family and each is the same identity read at one degree. At $d = 1$ the gradient of the deficit at $J_n/n$ is $\left(-2k/n + k \cdot k!/n^{k+1}\right)\mathbf{1}$, so $J_n/n$ is critical for every $(n, k)$. At $d = 2$ the Hessian has exactly two distinct eigenvalue parameters, $s_2 = k(k-1)/(n(n-1))$ and $t_2 = s_2^2 (k-2)!/n^{k-2}$, again for every $(n, k)$. And the coefficient of a degree-$d$ monomial of the deficit takes one of three values — $-2s_d + t_d$ if the cells form a partial permutation, $-s_d$ if they have distinct rows or distinct columns but not both, and 0 otherwise — so three numbers per degree describe the whole objective, whatever $k$ is. The closed forms used in §6.1 are the case $k = 3$ of that rule.

At $k = 2$ the sum has two terms and both are computable in closed form. Using $2\sigma_2(b) = (\sum b)^2 - \|R\|^2 - \|C\|^2 + \|b\|^2$ and $2e_2(R) = (\sum R)^2 - \|R\|^2$, and restricting to $\sum b = 0$, the deficit collapses to Equation 9, which is Theorem D. The identity was checked against the objective built from the 1992 definition in exact rational arithmetic at $n = 2, ..., 6$, four random $b$ per $n$ on the hyperplane, with no mismatches; the Lean proof of `subDittert_k2` goes through Equation 7 and does not use that check.

The same route does not reach $k = 3$. There the deficit is not a sum of squares on the hyperplane, and the certificate of §5 is needed.

# 4 The objective at $k = 3$, and how we know it is the right one

A certificate for the wrong polynomial is worthless and looks identical to a certificate for the right one, so the construction of the objective is checked before anything else. In centred coordinates $b = A - J_n/n$ put

$$F(b) = (2 - \gamma(n, k)) - [E_k(r) + E_k(c) - P_k(A)], \tag{15}$$

a polynomial of degree $k$ over $\mathbb{Q}$; the conjecture at $(n, k)$ is $F \geq 0$ on $K_n$. Five independent tests are run on the construction, all passing:

1. **$\sigma_k$ by two structurally different algorithms.** One enumerates all $\binom{n}{k}^2$ pairs of index sets and sums $k!$ products. The other never forms a submatrix at all: it uses $\mathrm{per}(A + xJ_n) = \sum_j x^{n-j}(n-j)!\,\sigma_j(A)$ in $\mathbb{Q}[x]$, with the order-$n$ permanent by Ryser inclusion–exclusion over column subsets. Forty random rational matrices agree, for $2 \leq n \leq 5$ and every $k$.
2. **Symbolic against from-scratch** evaluation at random rational points, at $(3, 2)$, $(4, 3)$, $(4, 4)$ and $(5, 3)$.
3. **A $k = n$ positive control.** The polynomial built here at $(4, 4)$ is compared coefficient by coefficient with the one from an independent Dittert pipeline: 1040 monomials each — that is the number carrying a non-zero coefficient in the degree-4 objective, out of the 4845 of degree at most 4 in the sixteen entries — and zero differing coefficients. That pipeline already underlies a verified certificate, so it is ground truth rather than a mirror.
4. The numerical trap of §4.1, explicitly separated.
5. $k = 1$ is confirmed to be an identity on $K_n$, and equality at $J_n/n$ is confirmed for every tested $(n, k)$.

## 4.1 Two coincidences that make the bound useless as a check

$2 - \gamma(4, 3) = 2 - 6/64$ and $2 - \gamma(4, 4) = 2 - 24/256$ are **both** $61/32$. The same collision recurs at $n = 5$: $2 - \gamma(5, 4) = 2 - \gamma(5, 5) = 1226/625$, which is the published Dittert value at $n = 5$. So seeing the right constant is **no evidence whatsoever** that the $k = 3$ or $k = 4$ code is doing what it claims.

The two cases are separated by comparing whole polynomials rather than constants. At $n = 4$ the $k = 3$ and $k = 4$ objectives have degrees 3 and 4; their supports together carry 1040 monomials — the $k = 3$ support, 552 monomials, sits inside the $k = 4$ one — and the two disagree at every one of the 1040. At $n = 5$ the same comparison spans 14005 monomials, degrees 4 and 5, again disagreeing at all of them.

# 5 The certificate

## 5.1 Form

In centred coordinates the ansatz is

$$F(b) = \sigma_0(b) + \sum_p \left(\frac{1}{n} + b_p\right) \sigma_p(b) + \lambda(b) \cdot \left(\sum_q b_q\right), \tag{16}$$

where $\sigma_0$ and every $\sigma_p$ is a sum of squares and $\lambda$ is a free polynomial. On $K_n$ the last term vanishes, since $\sum_q b_q = 0$ there, and every $1/n + b_p = A_p \geq 0$; so the right side is a sum of non-negative terms and $F \geq 0$ on $K_n$, which is the theorem.

The shape is Putinar's [2]: one sum-of-squares multiplier for each defining inequality $A_p \geq 0$, a free multiplier for the equality $\sum_q b_q = 0$, truncated at a fixed degree. No existence theorem is invoked. Putinar's applies to polynomials **strictly** positive on the set, and $F$ vanishes at $J_n/n$, in the relative interior of $K_n$; the representation is exhibited, not deduced, and the work is in getting it at the degree $F$ already has.

The toolset sits in a standard line: sum-of-squares relaxations of polynomial optimisation are Lasserre's hierarchy [13], and exploiting a symmetry group to block-diagonalise the Gram matrix is Gatermann–Parrilo [14]; both are used here in their plain forms. What is not taken from that line is the coefficient field: the programme is posed and solved over $\mathbb{Q}(n)$ — one symbolic solve whose output is a certificate for every $n$ at once, with positivity of the resulting rational functions decided by Sturm sequences rather than per-$n$ numerics. Optimisation uniform in the dimension is itself an active area — Levin and Chandrasekaran [15] develop any-dimensional polynomial optimisation via de Finetti theorems — and §13′s band law is this paper's measured answer, for one problem family, to when such uniformity survives at fixed certificate degree.

**Why the degree closes.** Here $\deg F = k$, not $n$. At $k = 3$ a Gram basis of degree 1 gives $\deg \sigma \leq 2$ and $\deg(\sigma_p b_p) \leq 3$, matching $\deg F$ exactly with no surplus top band to cancel. The Gram matrices are $n^2 \times n^2$. This is the structural reason the whole line $k = 3$ is accessible while a fixed $k = n$ is not: in the Dittert case the objective's degree grows with the dimension and the ansatz cannot follow it.

**Why rounding works.** The standing obstruction to exact rational rounding is that a tight bound forces a singular optimal Gram. Here the Hessian of $F$ at $J_n/n$ restricted to the tangent space $\{\sum X = 0\}$ is positive definite — at $(4, 3)$ its characteristic polynomial on the projected space is exactly $x(x - 1/16)^9(x - 29/16)^6$ over $\mathbb{Q}$, and the multiplicities $9 = (n-1)^2$ and $6 = 2(n-1)$ are forced by Schur's lemma, since the tangent space is $(V|1) \oplus (1|V) \oplus (V|V)$ with transposition fusing the first two. So there is no forced kernel inside the tangent space: centring at the extremiser and excluding the constant monomial removes the only degenerate direction.

## 5.2 Symmetry reduction, and the fact that its shape is fixed in $n$

The problem is invariant under $(S_n \times S_n) : \mathbb{Z}_2$ acting by row permutations, column permutations and transposition. Reducing Equation 16 by that group leaves, **at every $n$ alike**:

| **object** | **count, independent of $n$** |
|---|---|
| orbit constraint rows | $1 + 1 + 3 + 7 = 12$, of rank 11 over $\mathbb{Q}(n)$ |
| $\sigma_0$ variables | 3 |
| $\sigma_{11}$ variables | 11 |
| $\lambda$ variables | 5 |
| cone size | $n^2$ (the only thing that grows) |

Table 2: The symmetry-reduced program at $k = 3$ with a degree-1 Gram basis.

Nineteen unknowns and twelve equations, at $n = 4$, $n = 5$, $n = 6$ and beyond. This is representation stability, and here — unlike in the $k = n$ case, where the same stability is undercut by the growing degree — there is no counteracting obstruction. That observation is what makes a single symbolic solve plausible; the rest of the paper is the work of turning it into a proof.

One implementation point matters for reaching large $n$ at all. The transporter carrying the corner orbit to position $(i, j)$ is the explicit product of the row transposition $(0\,i)$ with the column transposition $(0\,j)$, at cost $O(n^2)$; enumerating the group instead costs $2(n!)^2$, which is $5.1 \times 10^7$ already at $n = 7$. The reduced system is provably unchanged, because two transporters differ by an element of the stabiliser of $(0, 0)$, which preserves each $\sigma_{11}$ orbit.

# 6 The uniform certificate

## 6.1 The constraint system in closed form, over $\mathbb{Q}(n)$

Both halves of the $12 \times 19$ system are **derived** as functions of $n$, not interpolated; "no fit is used anywhere" is a stronger statement than "the fit was validated".

**The right-hand side.** The coefficient in $F$ of an arbitrary monomial is closed form for all $n$ at once. Writing $d$ for the degree of the monomial and reading the two definitions directly,

```
[monomial] e_k(r)    = C(n-d, k-d)   if the d cells lie in distinct ROWS, else 0
[monomial] sigma_k   = C(n-d, k-d)^2 (k-d)! n^{-(k-d)}
                                     if the cells form a PARTIAL PERMUTATION, else 0
```

because within one product $r_i r_j r_k$ no index repeats, so two cells in a row cannot occur; and a term of a permanent is a bijection, so the cells must have distinct rows **and** distinct columns. This is the case $k = 3$ of the coefficient rule of §3. Checked against the fully expanded objective at $n = 4, 5, 6$: 969, 3276 and 9139 monomials, no mismatches. Those counts are of **every** monomial of degree at most 3 in the $n^2$ entries, absent ones included — the closed form has to return zero exactly where the expanded objective has no term, so the check is over the whole coefficient vector and not over the support. As a sanity check it gives the degree-one coefficient as $-6/n + 18/n^4$, which at $n = 4$ is $-183/128$, the gradient value computed independently.

**The orbit sizes.** By orbit–stabiliser, $|\text{orbit}(P)| = 2 \cdot [n]_r \cdot [n]_c / \ |\text{Stab } P|$ with falling factorials, and $|\text{Stab } P|$ found by brute force over at most $2 \cdot 3! \cdot 3! = 72$ relabellings. A multiset of at most three cells meets at most three rows and three columns, so every orbit size is a polynomial in $n$ of degree at most 6. Checked against brute-force enumeration at $n = 4, 5, 6, 7$, including the stabiliser case: no mismatches.

The structure that makes this work is that with a degree-1 basis the product $\text{basis}[u] \cdot \text{basis}[v]$ **is** the pair $\{u, v\}$, so the $\sigma_0$ variable orbits and the constraint-row orbits are the same objects; and canonicalisation is equivariant, so both incidence blocks are an orbit size times a 0/1 incidence.

**Cross-check against the trusted path.** The symbolic system was compared entry by entry, right-hand side included, with the system built by the original code path — the one that produced the already-verified fixed-$n$ certificates — at $n = 4, 5, 6$. No mismatches. The two share no logic: one does union–find over every monomial of the expanded objective, the other evaluates closed forms.

Row-reducing over the field $\mathbb{Q}(n)$ gives: 12 equations, 19 unknowns, rank 11, one dependent row, **consistent**, with an 8-dimensional affine family of solutions. So the linear half of the certificate exists at every $n$ at once.

## 6.2 Definiteness reduces to ten rational functions of $n$

What remains is to choose within that family so that both Gram matrices are positive definite for every $n \geq 4$. Both are block-diagonalised **in closed form** — a numerical basis at one $n$ could not settle all $n$.

**$\boldsymbol{\sigma_0}$.** Its Gram is $aI + bA_1 + cA_2$ in the Bose–Mesner algebra of the rook's graph $K_n \square K_n$, so it has exactly three eigenvalues:

$$\begin{aligned} \theta_0 &= a + 2(n-1)b + (n-1)^2 c \quad \text{multiplicity } 1, \\ \theta_1 &= a + (n-2)b - (n-1)c \quad \text{multiplicity } 2(n-1), \\ \theta_2 &= a - 2b + c \quad \text{multiplicity } (n-1)^2. \end{aligned} \tag{17}$$

$\boldsymbol{\sigma_{11}}$. With $V'$ the standard $(n-2)$-dimensional representation of $S_{n-1}$ one has $\mathbb{R}^{n^2} = 4(1|1) \oplus 2(V'|1) \oplus 2(1|V') \oplus (V'|V')$, and transposition fuses $(V'|1)$ with $(1|V')$ and splits the four-dimensional trivial multiplicity space as $3+1$. The blocks are a $3 \times 3$ (call it $A$), a $1 \times 1$ sign block $B$, a $2 \times 2$ block $C$ of multiplicity $2(n-2)$, and a $1 \times 1$ block $D$ of multiplicity $(n-2)^2$.

Both consistency checks pass: $\sum d(d+1)/2 = 6+1+3+1 = 11$, the number of orbit parameters, and $3+1+2 \cdot 2(n-2)+(n-2)^2 = n^2$. The decomposition was verified against a direct eigendecomposition of the assembled $n^2 \times n^2$ matrix on **random** orbit coefficients at $n = 4, ..., 8$ — full spectrum against the union of block spectra with predicted multiplicities, worst discrepancy $1.4 \times 10^{-13}$. Random coefficients matter here: a check run on the real certificate could pass by accident on a matrix carrying extra structure.

**Consequence.** Positive definiteness of both Grams, for all $n \geq 4$, is exactly the positivity of **ten** explicit rational functions of $n$: the three $\sigma_0$ eigenvalues, the three leading principal minors of $A$, the block $B$, the two leading minors of $C$, and the block $D$.

One of these ten is not independent of the rest: the block $D$ equals $\theta_2/n$ **exactly**, an identity of rational functions rather than a coincidence at sampled $n$ — the two share the same numerator and differ only by a factor of $n$ in the denominator. So positivity of $D$ follows immediately from positivity of $\theta_2$ together with $n > 0$, and the certificate rests on **nine** independent positivity facts, not ten; $D$ is carried along as a consequence rather than an additional condition. We report and certify all ten regardless — Sylvester's criterion asks for all ten directly, and the Sturm and Lean machinery below decides each on the same uniform footing — but a reader should not credit the certificate with ten independent facts where nine suffice.

These ten were validated first on the independently produced, already-verified certificates at $n = 4, 5, 6$: all ten strictly positive in each case.

## 6.3 The geometry of the feasible set

Choosing the eight free variables as functions of $n$ is the whole difficulty, and the shape of the difficulty is not the obvious one.

**The feasible set is unbounded, and the recession cone is exactly two-dimensional.** It is derived, not probed. A recession direction is a homogeneous solution with the sigmas still positive semidefinite; on the hyperplane $\sum b = 0$ every term of Equation 16 is non-negative on $K_n$, so each vanishes there, and a sum-of-squares quadratic form vanishing on a hyperplane has Gram $\mathbf{1}u^T + u\mathbf{1}^T$, which is positive semidefinite only for $u$ a non-negative multiple of $\mathbf{1}$. Hence the recession cone adds $c_0 J$ to the $\sigma_0$ Gram and $c_1 J$ to the $\sigma_{11}$ Gram, $c_i \geq 0$, with $\lambda$ determined. This was confirmed as an exact kernel element over $\mathbb{Q}(n)$ with a round trip through the free coordinates.

**The lineality space is larger, and it is what makes optima drift.** The quantities that actually have to be controlled need $G$ and $H$ positive semidefinite only on $\mathbf{1}^\perp$, which gives $G = \mathbf{1}u^T + u\mathbf{1}^T$ and $H = \mathbf{1}v^T + v\mathbf{1}^T$ with $u$ forced to a multiple of $\mathbf{1}$ by transitivity but $v$ free on the three stabiliser classes. Its four generators have an invertible $4 \times 4$ minor on the four free $\lambda$ columns, so the lineality space is exactly the $\lambda$ reparametrisation, and $\{f_{15} = f_{16} = f_{17} = f_{18} = 0\}$ is a transversal slice. On that slice the recession cone is trivial and **the feasible set is compact**. The essential design problem has **four** unknowns. Numerically, drift along the lineality space is plainly visible: the scaled free variable $f_6 n^3$

takes the values $149, 20.3, 9.69, 35311, 13.3, 83681, 6.66$ at $n = 20, 50, 100, 200, 500, 1000, 2000,$ which no curve in $n$ will follow.

**The essential set is a sliver, and that is why no fit can hold it.** In the scaled coordinates $\beta = f \cdot n^3$, four of the ten quantities are differences of terms of size $n^2$ or $n^3$ that must cancel down to $O(1)$. Working the cancellations out forces

$$\beta_6 - 2\beta_9 = O(n^{-2}), \quad \beta_{12} - (2\beta_9 - 1) = O(n^{-1}), \quad \beta_{11} - 2\beta_{12} - 2 = O(n^{-1}), \tag{18}$$

the last pinned to $O(n^{-3})$; asymptotically $\beta \to (2b, b, 4b, 2b-1)$, a one-parameter family with $1/2 < b < 1$, with the analytic centre at $b \approx 0.8486$. A least-squares fit whose residual is small against the diameter of the essential set is still far outside it, and a relation pinned to $O(n^{-3})$ is beyond the reach of curve fitting at any degree.

## 6.4 Adapted coordinates, and one exact elimination

Write the cancellations into the coordinates, so that they happen symbolically rather than numerically:

$$\beta_9 = b, \quad \beta_6 = 2b + \frac{x}{n^2}, \quad \beta_{12} = 2b - 1 + \frac{y}{n}, \quad \beta_{11} = 2\beta_{12} + 2 + \frac{z}{n}. \tag{19}$$

Then **solve for $z$ exactly over $\mathbb{Q}(n)$ from the equation $\theta_2 = D$**, rather than fitting it. This is the step that makes the difference. The two quantities are affine in $z$ with opposite-sign coefficients of size $n^2$, and the window between them has width $O(n^{-2})$; at $n = 10^6$ a numerical fit would need twelve correct digits merely to stay inside. With $z$ eliminated, the remaining parameters carry $\Theta(1)$ coefficients and an ordinary semidefinite program over a grid of $n$ finds them. The answer is simple:

$$b = 1, \quad x = 8 + \frac{20}{n}, \quad y = -2 - \frac{10}{n}, \tag{20}$$

$$z = \frac{6n^7 - 28n^6 + 41n^5 - 28n^4 + 48n^3 - 164n^2 + 208n - 80}{n^7 - 7n^6 + 19n^5 - 25n^4 + 16n^3 - 4n^2}. \tag{21}$$

The window $1/2 < b < 1$ of §6.3 and the choice $b = 1$ here are not in conflict, and the gap between them is worth naming. That window constrains the **limit** of the scaled quantities; the certificate is not the limit point. Its finite-$n$ corrections — $x/n^2$ and $y/n$ above, and $z$ solved exactly — are what carry the four tight quantities, and they are non-zero at every finite $n$. What the endpoint costs is margin, not sign: $\theta_2$ decays like $n^{-5}$. Positivity is then decided at every $n \geq 4$ by the Sturm step of §6.6, which acts on the exact rational functions and is indifferent to how small the margin becomes.

The transferable form of this step is short. In a family of certificates indexed by a parameter, identify the tight direction and eliminate it symbolically; fit only the slack ones. Quotienting the lineality space out is necessary — without it the optima drift and nothing converges — but it is not sufficient, because the compact essential set that remains can still be a sliver.

Two of the ten quantities cost nothing, because they are pure gauge. Moving along the lineality space adds $\mu n^2$ to $\theta_0$ and $sw^T + ws^T$ to the block $A$, with $s = (1, 2(n-1), (n-1)^2)$ and $w$ free in $\mathbb{R}^3$; choosing $w$ so that $A$ becomes $\operatorname{diag}(\gamma, T^T A T)$ in a basis $(e, T)$ with $e = (1, 0, 0)$, and taking $\gamma = 1/n^3$ and $\theta_0 = 1/n$, reduces $A \succ 0$ to $T^T A T \succ 0$, which is already one of the essential conditions. No new positivity requirement appears.

The resulting nineteen certificate variables are exact rational functions of $n$; they are recorded in full in the accompanying material and are not reproduced here. Three of them are $1/n^3$ exactly; the largest has numerator of degree 9 and denominator of degree 12.

## 6.5 Obstructions to simpler designs

Four routes that a reader would otherwise try are closed, each by an exact rational witness.

- **Linear programming for the design step is infeasible**, at ansatz degrees $D = 0, 1, 2, 3, 4$ (8 to 40 unknowns, 350–458 coefficient rows). The blocker is not the ansatz: plain diagonal dominance **fails on the already-verified certificates** at $n = 4, 5, 6$, in row 0 of both the $3 \times 3$ and the $2 \times 2$ block, so no LP built on dominance can be feasible. The cause is scaling — the isotypic vectors are unnormalised, so diagonal entries differ by orders of magnitude — and a rescaling does exist (both blocks are H-matrices at the verified certificates), but the weights that work are read off the certificate that the LP is trying to choose. The design step is **bilinear** in weights and certificate, not linear.
- **Least squares through analytic centres**, at degrees 1 to 7, fails off-grid at $n = 17$, $n = 71$ and $n = 811$ — always between grid points, always on a tight quantity.
- **Pinning four entries to constant targets** fails for all 210 four-subsets. The centre's own entries move by a factor of four between $n = 4$ and $n = \infty$, so constant targets pull the solution out of the set at one end or the other. Separately, the subset $\{\theta_2, D, C_{01}, T_{01}\}$ is exactly singular over $\mathbb{Q}(n)$; 197 of the 210 are non-singular.
- **Restricting $\boldsymbol{\lambda}$ to be a sum of squares** is sound, since $\lambda$ multiplies an equality constraint, but it removes only the two-dimensional recession cone and not the four-dimensional lineality space that causes the drift.

## 6.6 Sturm certification of all ten quantities

Substituting $n = m + 4$ turns each quantity into a ratio of polynomials in $m$ with the range becoming $m \geq 0$. The sign of each denominator on that range is **checked**, not assumed. The numerator's positivity on $[0, \infty)$ is then decided — decided, not estimated — by a Sturm chain on the squarefree part, comparing sign variations at 0 and at $+\infty$. There is no sufficiency gap in this step, which is precisely why the tempting shortcut "all coefficients non-negative in $m$" is only a heuristic: $m^2 - m + 1$ is positive everywhere and has a negative coefficient.

| **quantity** | **sqfree deg** | $V(0)$ | $V(\infty)$ | **roots in $(0, \infty)$** |
|---|---|---|---|---|
| `G0.theta0` | constant 1 | — | — | 0 |
| `G0.theta1` | 6 | 1 | 1 | 0 |
| `G0.theta2` | 4 | 1 | 1 | 0 |
| `H.A minor1` | constant 1 | — | — | 0 |
| `H.A minor2` | 5 | 1 | 1 | 0 |
| `H.A minor3` | 13 | 3 | 3 | 0 |
| `H.B` | 5 | 2 | 2 | 0 |
| `H.C minor1` | 5 | 1 | 1 | 0 |
| `H.C minor2` | 12 | 2 | 2 | 0 |
| `H.D` | 4 | 1 | 1 | 0 |

Table 3: All ten quantities, positive for every $n \geq 4$. Each row is a complete decision, not a sample: no roots in $(0, \infty)$ and a positive value at $m = 0$. `H.D` and `G0.theta2` share the same numerator — visible already in the matching squarefree degree 4 — because $D = \theta_2/n$ exactly (§6.2); we certify both rows independently regardless.

All ten denominators are sign-definite on $n \geq 4$, as are the denominators of all nineteen certificate variables, so nothing blows up at any dimension.

# 7 Verification

A certificate is worth what its checking is worth. The independent verifier **re-derives the objective from the 1992 definition** — $e_k$ of the row and column sums, and $\sigma_k$ as an explicit double sum of subpermanents over pairs of $k$-subsets — and uses none of the closed forms of §6.1, none of the block decomposition of §6.2 and none of the Sturm machinery of §6.6. The only shared code is the orbit bookkeeping that inflates nineteen numbers into two $n^2 \times n^2$ matrices, and that is exercised against stored certificates and against deliberate mutations.

At each $n$ it checks three things: that both Gram matrices are symmetric and **positive definite**, by exact rational $LDL^T$ on the **full** $n^2 \times n^2$ matrices rather than through the blocks; that the identity Equation 16 holds at random rational points; and that the constant is $2 - k!/n^k$. Positive definiteness of the single Gram at the corner gives it for every multiplier, since each $\sigma_p$'s Gram is a permutation conjugate of it.

| $n$ | **Gram** | $\sigma_0$ $LDL^T$ | $\sigma_{11}$ $LDL^T$ | **identity** |
|---:|:---:|:---:|:---:|:---|
| $4, 5, 6, 7$ | 16–49 | PD | PD | exact over $\mathbb{Q}$, 2 points each |
| 12 | $144 \times 144$ | PD, $9.17 \times 10^{-6}$ | PD, $7.77 \times 10^{-7}$ | exact over $\mathbb{Q}$, 2 points |
| 25 | $625 \times 625$ | PD, $1.45 \times 10^{-7}$ | PD, $5.80 \times 10^{-9}$ | $\mathbb{F}_p$, 3 primes, 2 points |

Table 4: Independent verification of the uniform certificate. Entries under $LDL^T$ are the smallest exact pivot.

Because the definiteness check runs on the unblocked matrices, it is an independent confirmation of the block theory of §6.2 as well as of the numbers.

**Why $\mathbb{F}_p$ at $n = 25$, and why that is still exact.** The rational evaluation would need about $2.4 \times 10^8$ `Fraction` multiplications per point. The same computation modulo a prime is exact arithmetic — not floating point — and vectorises: with $p < 2^{20}$ every intermediate stays inside 64-bit integers, so the 625 quadratic forms become 625 matrix–vector products. Run at three unrelated primes near $2^{20}$, a surviving discrepancy would have to be divisible by all three.

**Positive control.** The same verifier is run on the independently produced, already-verified **stored** certificates at $n = 4$ and $n = 5$, using their own orbit data rather than any mapping of ours, and accepts them. If that had failed the verifier would have been the thing at fault.

**Negative controls, four of them.** Three single-variable perturbations of the uniform certificate at $n = 4$ — variables 0, 5 and 16, each moved by $10^{-6}$ — are rejected over $\mathbb{Q}$; a perturbation at $n = 6$ is rejected in $\mathbb{F}_p$ at all three primes. A verifier that never rejects proves nothing.

## 7.1 The fixed-dimension certificates

**Support layer: exact rational certificates, accepted by a standalone verifier. These are not Lean-checked, and no theorem of this paper depends on them.**

Five certificates at fixed dimensions were produced independently of the uniform construction, each with its own numerical solve and its own rounding. Each establishes the stated bound on the stated $K_n$, with equality only at $J_n/n$: $\Phi_3 \leq 16/9$ on $K_3$; $\Phi_3 \leq 61/32$ on $K_4$; $\Phi_3 \leq 244/125$ on $K_5$; $\Phi_3 \leq 71/36$ on $K_6$; and, at a second value of $k$ in the intermediate range, $\Phi_4 \leq 1226/625$ on $K_5$.

The four $k = 3$ cases are covered by Theorem A and Corollary B, equality cases included. They are retained because they are the anchors against which the uniform certificate was checked, and because they were rounded independently of it. The case $(5, 4)$ is not reached by the uniform construction.

| $(n,k)$ | **bound** | **basis deg** | **monomials in $F$** | **cone** | **certificate** |
|---|---|---|---|---|---|
| $(3,3)$ | 16/9 | 2 | 93 | 54 | 314 rationals |
| $(4,3)$ | 61/32 | 1 | 552 | 16 | 19 rationals |
| $(5,3)$ | 244/125 | 1 | 2225 | 25 | 19 rationals |
| $(6,3)$ | 71/36 | 1 | 6906 | 36 | 19 rationals |
| $(5,4)$ | 1226/625 | 2 | 7875 | 350 | 440 rationals |

Table 5: The five fixed-dimension certificates. Every one is exact rational data; the identity is confirmed by **full coefficient comparison**, not by sampling.

Smallest exact $LDL^T$ pivots, as the standalone verifier reports them — first the $\sigma_0$ Gram, then the smallest over **all** $n^2$ multiplier Grams: $8.24 \times 10^{-3}$ and $8.36 \times 10^{-3}$ at $(3,3)$; $6.94 \times 10^{-3}$ and $6.65 \times 10^{-3}$ at $(4,3)$; $2.02 \times 10^{-3}$ and $1.93 \times 10^{-3}$ at $(5,3)$; $2.72 \times 10^{-4}$ and $6.48 \times 10^{-4}$ at $(6,3)$; $5.83 \times 10^{-4}$ and $5.87 \times 10^{-4}$ at $(5,4)$. Mutation tests perturbing a single coefficient by $10^{-20}$ are rejected in every case.

The second number needs its matrix named, because two are in circulation. The $n^2$ multiplier Grams are permutation conjugates of one another and so share a spectrum, but an $LDL^T$ pivot depends on the order of the indices, and the smallest pivot is not constant across the orbit. The design pipeline factors the Gram at the corner $p = (0,0)$ and reports that one — $7.32 \times 10^{-3}$ at $(4,3)$, $7.11 \times 10^{-4}$ at $(6,3)$ — where the verifier factors all $n^2$ and reports the minimum. Both decide the same question, and the verifier's figure is the conservative one.

All five cases pass the standalone verifier in full — six checks each, standard library only, no shared code with the pipeline. In case $(5,4)$ the exported data writes all 26 Gram matrices out densely so that the verifier needs no group theory, so it runs 26 exact rational factorisations of size 350 where only two are distinct. That redundancy is deliberate: exploiting the conjugacy would mean either trusting the conjugating permutation or verifying it, which is exactly the group theory the dense export exists to keep out of the verifier. No matrix is skipped.

$(5,4)$ also needed the block-diagonalisation to be solvable at all: at cone size 350 a monolithic interior-point solve needs of order 30 GB per cone. Blocked, $\sigma_0$ gives cones $[5,5,4,2,2,1,1,1,1,1]$ and $\sigma_{11}$ gives $[16,14,10,7,4,4,3,2,1,1,1]$ — identical to the blocks already recorded for the Dittert case at $n = 5$ — with largest cone 16.

# 8 Confinement, for every $k$

Theorem E is recalled from [6] and transferred to all $k$; nothing in this section is claimed as new mathematics. It is included because it is uniform in $k$ where the certificate of §5 is not, and because it is short enough to formalise once for the whole family.

The proof needs no expansion of the objective. Split the deficit as

$$(2 - \gamma(n,k)) - \Phi_k(A) = [1 - E_k(r)] + [1 - E_k(c)] + [P_k(A) - \gamma(n,k)], \tag{22}$$

where the third bracket is bounded below by $-\gamma(n,k)$ because $P_k$ of a non-negative matrix is non-negative. For the first two brackets, Maclaurin's inequality in the telescoped form of Theorem F gives $E_k(r) \le E_2(r)$ on the simplex, and $E_2$ is exactly computable:

$$E_2(r) = 1 - \frac{\sum_i (r_i - 1)^2}{n(n-1)} \quad \text{whenever} \sum_i r_i = n, \tag{23}$$

which is `SubDittertK2.E_two_eq` and needs no positivity. Together these give Theorem E, and its contrapositive is the confinement statement: a violator of the Cheon–Hwang bound has its line sums within $(n-1)k!/n^{k-1}$ of $\mathbf{1}$ in squared $\ell^2$ distance, a neighbourhood that shrinks in $n$ for every fixed $k \geq 2$.

The hypothesis $E_k \leq E_2$ is stated in Lean as a named predicate, `SubDittertM.MaclaurinBound`, and `SubDittertMaclaurin.maclaurinBound_holds` discharges it for every $2 \leq k \leq n$; `theoremM'` and `confinement'` are the resulting unconditional statements. At $k = 2$ the predicate holds with equality, so the conclusion is not vacuous there.

### 8.1 Newton's and Maclaurin's inequalities in Lean

Theorem F is classical, and the contribution is the formalisation. The route avoids the reversal of polynomials that the textbook proof uses and whose root theory Mathlib does not carry. Writing $Q_i = \mathrm{coeff}_i/\binom{m}{i}$ for a polynomial of degree $m$, one has $Q_i' = m \cdot Q_{i+1}$, so Newton's inequality at index $i$ for $p$ is Newton's inequality at index $i-1$ for $p'$, and $p'$ is real-rooted whenever $p$ is. That reduces every index $i \geq 2$ to a lower degree. Differentiation shifts the index upward, so $i = 1$ is never reached by the induction; it is not a base case at one degree but has to be proved directly at every degree, and it is where real-rootedness is consumed. The ingredients are Rolle's theorem for real-rooted polynomials, Vieta's formulas, the reciprocal identity $e_{n-j}(v) = \left(\prod_i v_i\right) \cdot e_j(v^{-1})$, and Chebyshev's sum inequality.

A second, independent Lean proof of Theorem F exists, written without sight of the first and by a different route — Mathlib's `coeff_eq_esymm_roots_of_card` with an induction on the degree, against an induction on the index — and its statements agree with those above. It is stored as verification evidence outside the build, in its own namespace (`NewtonCrosscheck`); agreement between two independent proofs is evidence that the statements are the intended ones, which is the one thing a kernel cannot check.

## 9 Formalisation

### 9.1 The verification standard

The development elaborates with no `sorry`, and no declaration in it depends on `sorryAx`. `native_decide` is used nowhere. Every declaration this paper cites in support of a stated result carries a `#print axioms` command, and each returns exactly `[propext, Classical.choice, Quot.sound]` — with one exception, `NewtonIneq.choose_mul_sub`, a statement about binomial coefficients, which returns the strict subset `[propext, Quot.sound]`.

All counts here are of the Lean sources as committed at `32c811e`, the commit at which they were last changed. The eight files carrying the results of this section contain 511 axiom-audited declarations: 377 in `SubDittertK3.lean` and 7 in `RookSum.lean`, which together prove Theorem A; 21 in `SubDittertUniversal.lean` (Theorem C), 17 in `SubDittertK2.lean` (Theorem D), 37 in `SubDittertLinear.lean`, 13 in `SubDittertM.lean` and 6 in `SubDittertMaclaurin.lean` (Theorem E), and 33 in `NewtonInequalities.lean` (Theorem F). In `SubDittertK3.lean` the 377 are **every** named declaration in the file, so nothing in the file carrying Theorem A is left unaudited. The independent cross-check of §8.1 adds 12 more and is deliberately outside the build.

Theorem G's formalised cells rest on further files, counted at their own commits and held to the same standard — `StabilityK3.lean` (33 audit lines) and `TverbergStability.lean` (21) at `1507013`, `LayerIdentity.lean` (28) at `27bda3f`, `SigmaFour.lean` (46) at `365e44e`, `StabilityK4.lean` (14) at `944b517`, and the partial `StabilityK5Atoms.lean` (15) at `f2bdc7f`. §10.10 is their record, and the verifier of §10.9 checks that record against those commits mechanically.

In the other files the audits cover the results and their supporting lemmas rather than every private definition. That is not a gap: `#print axioms` reports the axioms of the **transitive closure** of a proof term, so auditing a theorem audits every definition and lemma the theorem rests on. No file contains a `sorry`, so nothing in any of them can acquire `sorryAx` from within; and an unaudited declaration on which no stated result depends cannot affect a stated result.

The development is built against Lean 4.14.0 and Mathlib `v4.14.0` [16]. `NewtonInequalities.lean` imports Mathlib and nothing else — no permanent, no $K_n$ — because the results it proves are library material and the boundary should stay visible.

What the kernel cannot check is the translation of the 1992 statement into Lean's definitions. That is what §9.2 item 2 defends, and it is the only place left where a human error would survive the machine.

## 9.2 Theorem A, in the order it is built

**`subDittert_k3_full` is Theorem A, in all three parts.** For every $n \geq 4$ and every $A$ with non-negative entries summing to $n$, it proves $E_3(r) + E_3(c) - P_3(A) \leq 2 - 6/n^3$; that equality holds if and only if $A = J_n/n$; and the stability bound Equation 5 with the explicit $\theta_2(n)$ — with every definition in those statements built from scratch inside Lean.

1. **The statement.** `Kn`, `rowSum`, `colSum`, `esym`, `subPerm`, `sigmaK`, `E`, `P` and `Phi` are defined from scratch, with $\sigma_k$ a double sum over `Finset.powersetCard k univ` of the permanent of the submatrix on rows $S$ and columns $T$. The final statement `subDittert_k3` is written out in the 1992 notation.
2. **Validation of the definitions**, which matters more than it looks. `Phi_uniform` proves $\Phi_k(J_n/n) = 2 - k!/n^k$ for **every** $k \leq n$, so the functional as formalised is tight at the conjectured maximiser exactly where the conjecture says it is; instances at $n = 3, 4, 5$ specialise it to $2 - 6/n^3$. `sigmaK_rankOne` proves $\sigma_k(xy^T) = k! \cdot e_k(x) \cdot e_k(y)$, which is the check that rows are read from $S$ and columns from $T$: a definition that reads both indices off the same subset satisfies a different identity, and a symmetric test matrix cannot detect that error, whereas this does. A concrete $\sigma_3 = 450$ on an explicit $3 \times 3$ matrix is checked against a hand value.
3. **The combinatorial half, closed for all $n$ at once.** `RookSum.lean` proves a closed form for $\sigma_3$ directly from the subpermanent definition: the bridge is a bijection between $k$-subsets paired with a permutation of them and injections Fin $k \to$ Fin $n$, stated and proved general in $k$, specialised to $k = 3$ by a three-index inclusion–exclusion sieve. On top of that, `obj_eq_objPoly` proves that the objective $(2 - 6/n^3) - \Phi_3(A)$ equals an explicit polynomial `objPoly n A` in the entries of $A$ and its row and column sums, for every $n \geq 3$. Together these eliminate `Matrix.permanent`, `subPerm`, `sigmaK`, `esym` and every sum over `Finset.powersetCard` from everything downstream: what remains of the proof mentions none of them. `objPoly` is cross-validated outside Lean against the from-the-1992-definition objective of §7, exactly over $\mathbb{Q}$, at random rational points for $n = 4, 5, 6, 7$, at $J_n/n$ where it must vanish, and against a mutation control.
4. **The ten positivity facts.** `certPositive_of_four_le` proves all ten quantities of §6.6 positive for every **real** $n \geq 4$ — real, not merely natural, which is strictly stronger and no harder here. Sturm is not needed inside Lean: after the substitution $n = m + 4$ all twenty polynomials involved (ten numerators, ten denominators) have non-negative coefficients and a positive constant term, so each proof is one `ring` and one `linarith`. As noted in §6.2, one of the ten — `H.D` — is not independent ($D = \theta_2/n$ exactly), so this proves nine independent facts and one immediate consequence of them; Lean proves all ten on the same uniform footing regardless, since the substitution argument costs nothing extra to repeat.

5. **Both Gram families, and the $\sigma_0$ Gram positive definite.** `G0_posSemidef` and `Hm_posSemidef` prove semidefiniteness for every $n \geq 4$, with no spectral theorem in either, and `G0_posDef` upgrades the first to **definiteness** — which is what the equality case needs. The $\sigma_0$ form splits into four manifestly non-negative pieces, one of them $\theta_2(n) \cdot \sum_u b_u^2$; since $\theta_2(n) > 0$ is already among the ten facts, that piece alone is strictly positive off the origin, so definiteness costs ten further lines and no new mathematics. The eigenvalue multiplicities of §6.2 are not needed for it. For a multiplier Gram the coordinates are the residuals about the corner's own row and column; in those the form breaks into blocks $A$, $B$, $C$, $D$ — the isotypic blocks of §6.2, in this basis rather than the one §6.2 normalises them in — and each is bounded below by a completion of squares, Lagrange's identity in two variables and Sylvester's in three. Seven positive quantities drive that, and they are the leading principal minors of those four blocks **times the constants** $1, 2, 4, 2, 1, 4, 1$. The constants do not depend on $n$, so positivity is the same question either way, but a bridge lemma stated without them is false. They are checked outside Lean as identities of rational functions, and the change of coordinates exactly over $\mathbb{Q}$ at $n = 4, 5, 6$ at three corners, with a mutation control.
6. **The Positivstellensatz identity.** `certificate_identity` proves Equation 16 itself, for every $n \geq 4$ and every $A \in K_n$, with $G_0$ and the family $H_p$ the explicit Grams above. Both sums over corners are expanded in ten global invariants of the centred coordinates. Every coefficient in that expansion was emitted rather than typed, and checked outside Lean as an identity of rational functions of $n$ by full coefficient comparison rather than by sampling.
7. **The deduction.** `subDittert_k3_of_certificate` derives the bound from the certificate through the bridge lemma: non-negativity of the quadratic forms from `PosSemidef`, non-negativity of the entries of $A$ from membership of $K_n$, then `linarith`. The $\lambda(b) \cdot \left(\sum_q b_q\right)$ term does not appear, because that sum vanishes on $K_n$.
8. **The equality case and the stability bound**, both for every $n \geq 4$. `subDittert_k3_bound_and_uniqueness` is the bridge lemma's packaged form `certificate_bound_and_uniqueness` applied to the certificate's own data; its separation hypothesis is `eq_uniform_of_centre_eq_zero`, which is the remark that the centred coordinates are $b_p = A_p - 1/n$ entrywise, so their vanishing **is** $A = J_n/n$. `subDittert_k3_stability` keeps the $\theta_2(n) \cdot \sum_u b_u^2$ piece instead of discarding it and so proves Equation 5, with `subDittert_k3_stability_explicit` the same statement with the constant written out; `eq_uniform_of_Phi_eq_of_stability` then re-derives uniqueness from stability alone, without the positive-definite bridge, so the equality case stands on two independent Lean proofs. `subDittert_k3_full` states all three parts of Theorem A as one theorem in the 1992 notation.

**A normalisation caveat, stated exactly**, on the ten quantities of the fourth item — a separate matter from the seven constants of the fifth. Each numerator and denominator in the Lean file is separately cleared to primitive integer coefficients, and the pair is sign-flipped together where the denominator is negative on $n \geq 4$. Both operations scale by a positive rational, so each Lean quantity is a constant **positive** rational multiple of the corresponding quantity in the Sturm pipeline; the multiple is 1 except for `H.A minor2` and `H.B` (both 1/2) and `H.C minor2` (4). **These four numbers are an artefact of two independent choices of primitive-integer clearing, one on each side, and carry no mathematical content of their own**; a different but equally valid clearing convention on either side would change 1/2 and 4 to other positive rationals without changing anything else. This was cross-checked at $n = 4, 5, 7, 13, 101$: the ratio is constant and positive in every case, so positivity of one is positivity of the other, which is the only fact that matters.

## 9.3 Theorems C to F in Lean

The uniform half of the development is independent of the certificate and much shorter.

`universal_identity` (Theorem C) is proved for every $n$ and every $k \leq n$ from two coefficient rules. The $e_k$ rule is the centre identity `esym_one_add` of `SubDittertLinear.lean`; the $\sigma_k$ rule, `sigmaK_one_add`, is proved through a rook bridge valid at every $k$ — summing over pairs of injections Fin $k \to$ Fin $n$ is $k!$ times the subpermanent sum — with the fibres of the restriction map counted by Mathlib's `Equiv.sumEmbeddingEquivSigmaEmbeddingRestricted`. One supporting lemma, `choose_subset_of_subset`, is absent from Mathlib and is a candidate for upstreaming.

`subDittert_k2` (Theorem D) instantiates `universal_identity` at $k = 2$, closes $\sigma_2$ and $e_2$ in closed form by the $k = 2$ sieve, and concludes through the same bridge lemma the $k = 3$ proof uses. There is no Gram matrix in it.

`theoremM'` and `confinement'` (Theorem E) are `theoremM` and `confinement` with the hypothesis `MaclaurinBound` discharged by `maclaurinBound_holds`. The attribution to [6] is the first section of the header of the file that states them.

`newtonAt_all`, `newton_esymF` and `pnorm_le_two` (Theorem F) are proved in a file that imports Mathlib alone.

## 9.4 Equality and stability

For each of the five fixed-dimension certificates of §7.1, equality holds **only** at $J_n/n$. The argument is one line: the $\sigma_0$ Gram is positive **definite** by exact $LDL^T$, and its monomial basis excludes the constant, so $\sigma_0(b) = 0$ forces $b = 0$, while every other term of Equation 16 is non-negative on $K_n$.

For every $n \geq 4$ the same conclusion is Lean-proved, and it needs no part of the block-diagonalisation of §6.2. The Bose–Mesner form of the $\sigma_0$ Gram makes its quadratic form

$$\sigma_0(b) = c_0^{\mathrm{tot}} \left( \sum_u b_u \right)^2 + c_0^{\mathrm{line}} \left( \sum_i R_i^2 + \sum_j C_j^2 \right) + \theta_2(n) \sum_u b_u^2 \tag{24}$$

identically in $b$, with all three coefficients positive on $n \geq 4$; the identity is `quadForm_G0`. The last term alone is strictly positive unless every $b_u$ vanishes, so definiteness follows from $\theta_2(n) > 0$, one of the ten facts of §6.6, by an argument that mentions neither eigenvalues nor their multiplicities. The spectral decomposition is the route by which the coefficients were found; the finished identity does not depend on that route.

Keeping the $\theta_2$ term of Equation 24 rather than discarding it costs nothing and gives Equation 5, which is strictly stronger than the equality case. The constant is **not** claimed optimal. Both sides of Equation 5 were evaluated exactly over $\mathbb{Q}$ at $n = 4, 5, 6$ and the slack ratio measured. Along directions where the row and column sums of $A - J_n/n$ all vanish, the first two terms of Equation 24 vanish identically, only the multiplier half of the certificate is being discarded, and the ratio is constant in the size of the perturbation — 2.88, 4.34, 5.70 at $n = 4, 5, 6$. So $\theta_2(n)$ is within a factor of about three to six of the best constant in the directions where it is closest to tight, and further off in general directions. Sharpening it would mean retaining part of the multiplier half, which we have not attempted.

## 9.5 What is not formalised

One thing is not: the block-diagonalisation of §6.2 as an **equivalence** — the Bose–Mesner eigenvalues with their multiplicities, the isotypic decomposition, and with them the statement that positive definiteness of the two Grams is exactly the positivity of the ten rational functions. Lean proves the implication the theorem needs and not the converse, and it proves it by explicit algebra rather than by that route, so the reduction of §6.2 is corroborated by the Lean development without being verified by it. The two are checked against each other outside Lean: the unblocked $n^2 \times n^2$ matrices are factored directly at six dimensions in §7.

The fixed-dimension certificates of §7.1 are also unformalised, as §1.2 records.

## 10 A stability form of the Tverberg–Friedland theorem

Tverberg conjectured in 1963 [17], and Friedland proved in 1982 [18], that $\sigma_k$ attains its minimum on the doubly stochastic polytope $\Omega_n$ only at the barycentre $J_n/n$; at $k = n$ the statement is the generalised van der Waerden conjecture, whose $k = n$, $\sigma_n =$ per case was settled by Egorychev [20] and Falikman [21]. Those theorems give strictness but no rate. Theorem G is the quantitative form: for $2 \le k \le 5$, on the per-$k$ ranges stated with it, the deficit is bounded below by an explicit positive multiple of the squared Frobenius distance to the barycentre. The excluded cells are finite in number for each $k$:

$$(k, n) \in \{(3, 3)\} \cup \{(4, n) : 4 \le n \le 7\} \cup \{(5, n) : 5 \le n \le 13\}. \tag{25}$$

One of them is a genuine obstruction rather than a gap in the argument: at $n = k = 3$ the inequality with this constant is false, and §10.7 exhibits the witness. For the remaining excluded cells the argument below does not reach, and no claim is made either way.

Everything in this part is confined to the doubly stochastic polytope. The estimates below use four a-priori facts (§10.4) that hold on $\Omega_n$ and fail on the larger set $K_n$; nothing in this part should be read as a statement about $K_n$. The transfer to a collar around the Birkhoff polytope, and the degradation it costs, is the business of §11 — that is where Theorem H consumes this part.

### 10.1 Setting and notation

Fix $n \ge 2$ and let $A \in \Omega_n$. Put

$$B = A - J_n/n, \quad Q = \|B\|_F^2 = \sum_{i,j} b_{ij}^2. \tag{26}$$

Since $A$ is doubly stochastic, every row and every column of $B$ sums to zero; we refer to this as the **centred slice**. Write

$$q_i = \sum_j b_{ij}^2, \quad q'_j = \sum_i b_{ij}^2, \quad M = \max\Big(\max_i q_i, \max_j q'_j\Big), \quad \beta = 1 - \frac{1}{n}, \tag{27}$$

so that $Q = \sum_i q_i = \sum_j q'_j$. For $r \ge 2$ set $p_r = \sum_{i,j} b_{ij}^r$, and

$$Y_R = \sum_i q_i^2, \quad Y_C = \sum_j q_j'^2, \quad Z = \left\|B^T B\right\|_F^2. \tag{28}$$

Four further invariants appear at degree five. With $f_3(i) = \sum_j b_{ij}^3$ and $g_3(j) = \sum_i b_{ij}^3$, and with $q, q'$ read as vectors,

$$\Gamma_a = q^T B\, q', \quad \Gamma_b = \sum_{i,j} b_{ij}^2 (BB^T B)_{ij}, \quad \Gamma_c = \sum_i q_i f_3(i), \quad \Gamma'_c = \sum_j q'_j g_3(j). \tag{29}$$

Finally, for $0 \le m \le k$, $s_m = [k]_m/[n]_m$ and $t_m = s_m^2 \cdot (k - m)!/n^{k-m}$ — the coefficients of Theorem C, in the index letter $m$ of the layer they weight. Two facts about them are used repeatedly. First,

$$t_2 = \frac{k(k-1)\,k!}{n^k (n-1)^2}, \quad \text{so} \quad c(n, k) = t_2/4. \tag{30}$$

Second, the ratio law

$$\frac{t_{m+1}}{t_m} = \frac{n(k-m)}{(n-m)^2}. \tag{31}$$

Both are immediate from the definitions.

## 10.2 The layer identity

**Lemma S1.** For every $A \in \Omega_n$ and every $1 \le k \le n$,

$$\frac{\sigma_k(A)}{\binom{n}{k}^2} - \frac{k!}{n^k} = \sum_{m=2}^{k} t_m\, \sigma_m(B), \quad B = A - J_n/n. \tag{32}$$

Lean: `layer_identity` in `LayerIdentity.lean` (commit `27bda3f`), kernel-checked at **every** $k$, under hypotheses weaker than this part needs: see §10.10.

Lemma S1 is the $P_k$ half of Theorem C's decomposition Equation 7 read on the centred slice, where the $e_d$ half vanishes; the proof below is self-contained and is the one the Lean development formalises.

**Proof.** Expanding a permanent of a sum, for index sets $\alpha, \beta$ of size $k$,

$$\operatorname{per}((X+Y)[\alpha|\beta]) = \sum_{S\subseteq\alpha, T\subseteq\beta,\ |S|=|T|} \operatorname{per}(X[S|T]) \operatorname{per}(Y[\alpha \setminus S \mid \beta \setminus T]). \tag{33}$$

Take $X = J_n/n$ and $Y = B$, so $\operatorname{per}(X[S|T]) = d!/n^d$ for $|S| = |T| = d$. Summing over all $\alpha, \beta$ and writing $m = k - d$ for the size of the $B$-part, each pair $(U, V)$ with $|U| = |V| = m$ is completed to a pair $(\alpha, \beta)$ by choosing $S$ disjoint from $U$ and $T$ disjoint from $V$, which can be done in $\binom{n-m}{k-m}^2$ ways. Hence

$$\sigma_k(A) = \sum_{m=0}^{k} \binom{n-m}{k-m}^2 \frac{(k-m)!}{n^{k-m}} \sigma_m(B). \tag{34}$$

Since $\binom{n-m}{k-m}/\binom{n}{k} = s_m$, dividing by $\binom{n}{k}^2$ turns the coefficient into $t_m$. The term $m = 0$ contributes $t_0 = k!/n^k$, and the term $m = 1$ vanishes because $\sigma_1(B) = \sum_{i,j} b_{ij} = 0$. ∎

Write $F = \sum_{m=2}^{k} t_m \sigma_m(B)$ for the left side of Equation 32. Since $t_m > 0$ for all $m \le k$, Theorem G is equivalent to

$$\sum_{m=3}^{k} t_m\, \sigma_m(B) \ge -\frac{1}{4} t_2\, Q, \tag{35}$$

because $\sigma_2(B) = Q/2$ on the centred slice (§10.3), so that $t_2 \sigma_2(B) = \frac{1}{2} t_2 Q$ and the two halves of $\frac{1}{2} t_2 Q$ split between the claimed bound and the allowance in Equation 35. It is Equation 35 that we prove.

## 10.3 The core expansions

On the centred slice each $\sigma_m(B)$ collapses onto a small set of invariants. Expanding $\sigma_m$ by Möbius inversion over ordered pairs of set partitions of $[m]$, a partition with a singleton block contributes a factor that is a row sum or a column sum of $B$, hence zero. Equivalently, only those orbit invariants survive whose associated bipartite multigraph has minimum degree at least two.

This expansion, together with the reductions at degrees two, three and four below, is due to McCullagh [19, §3]; the degree-five reduction is a specialisation of the general formula given there. In the notation of §10.1, for $B$ with vanishing row and column sums,

$$\sigma_2(B) = \frac{1}{2} Q, \tag{36}$$

$$\sigma_3(B) = \frac{2}{3} p_3, \tag{37}$$

$$\sigma_4(B) = \frac{3}{2} p_4 + \frac{1}{8} Q^2 + \frac{1}{4} Z - \frac{3}{4} Y_R - \frac{3}{4} Y_C, \tag{38}$$

$$\sigma_5(B) = \frac{24}{5}p_5 + \frac{1}{3}Q\,p_3 + \Gamma_a + 2\Gamma_b - 4\Gamma_c - 4\Gamma_c'. \tag{39}$$

The coefficient of $p_m$ in $\sigma_m$ is $(m-1)!/m$, the only contributing partition pair being the maximal one in each coordinate; this accounts for $\frac{1}{2}, \frac{2}{3}, \frac{3}{2}, \frac{24}{5}$ in equations 36–39.

Lean: the $\sigma_4$ expansion Equation 38 is kernel-checked at **every** centred $B$ — `sigma_four_centred` in `SigmaFour.lean` (commit `365e44e`), stated with both marginal hypotheses and consuming both, so the identity holds wherever the row and column sums vanish, not only on $\Omega_n - J_n/n$.

## 10.4 Four a-priori facts on the slice

**Lemma S2.** Let $A \in \Omega_n$ and $B = A - J_n/n$. Then

**(F1)** $-\frac{1}{n} \le b_{ij} \le 1 - \frac{1}{n}$ for all $i, j$;

**(F2)** $M \le 1 - \frac{1}{n}$;

**(F3)** $Q \le n - 1$;

**(F4)** $\|B\|_{\mathrm{op}} \le 1$, and consequently $Z \le Q$.

**Proof.** (F1) Each entry of $A$ lies in $[0,1]$, since $0 \le A_{ij}$ and $A_{ij}$ is at most the sum of its row, which is 1. Subtract $1/n$.

(F2) $q_i = \sum_j A_{ij}^2 - \frac{2}{n}\sum_j A_{ij} + n \cdot \frac{1}{n^2} = \sum_j A_{ij}^2 - \frac{1}{n}$, and $\sum_j A_{ij}^2 \le \left(\max_j A_{ij}\right)\sum_j A_{ij} \le 1$ by (F1). The same computation applies to columns.

(F3) Sum (F2) over $i$: $Q = \sum_i q_i \le n\left(1 - \frac{1}{n}\right) = n - 1$. Equality holds at every permutation matrix.

(F4) Let $u = n^{-1/2}\mathbf{1}$. Then $Au = u$ and $(J_n/n)u = u$, so $Bu = 0$. For $x \perp u$ we have $(J_n/n)x = 0$, hence $Bx = Ax$. By Birkhoff's theorem $A$ is a convex combination of permutation matrices, each an isometry, so $\|Ax\| \le \|x\|$. Decomposing an arbitrary $x$ as $\alpha u + x^\perp$ gives $Bx = Bx^\perp = Ax^\perp$ and therefore $\|Bx\| \le \|x^\perp\| \le \|x\|$. For the consequence, $Z = \|B^T B\|_F^2 \le \|B^T\|_{\mathrm{op}}^2 \|B\|_F^2 \le Q$. ∎

All four are attained at the permutation matrices, so none can be improved by a constant factor. Birkhoff's theorem is where double stochasticity is consumed; (F4) is the fact that fails first off the slice, and §11.6 is what replaces the whole lemma on the collar.

## 10.5 Per-invariant estimates

The estimates below are one-sided by design. Each invariant is bounded only on the side that the sign of its coefficient in equations 37–39 requires, and the asymmetry of the range in (F1) makes the required side the cheap one. This is where the entry bound earns its keep: for $b$ in $\left[-\frac{1}{n}, 1 - \frac{1}{n}\right]$,

$$b^3 \ge -\frac{1}{n}b^2 \quad \text{and} \quad b^3 \le \left(1 - \frac{1}{n}\right) b^2, \tag{40}$$

the first because $b^3 \ge 0$ when $b \ge 0$ and $b^3 = b \cdot b^2 \ge -\frac{1}{n}b^2$ when $b < 0$, the second symmetrically. Summing the first over all entries gives $p_3 \ge -Q/n$, a factor $n$ stronger than the two-sided bound $|p_3| \le \beta Q$, which would be too weak for what follows.

**Lemma S3.** On the centred slice:

| estimate | attained |
|---|---|
| (a) $p_3 \geq -Q/n$ | no |
| (b) $p_5 \geq -Q/n^3$ | no |
| (c) $p_4 \leq \beta^2 Q$ | no |
| (d) $Y_R \leq MQ$ and $Y_C \leq MQ$ | **yes** |
| (e) $\Gamma_c \leq \beta MQ$ and $\Gamma_c' \leq \beta MQ$ | no |
| (f) $\lvert\Gamma_a\rvert \leq MQ$ | no |
| (g) $\lvert\Gamma_b\rvert \leq \beta Q$ | no |
| (h) $Q\,p_3 \geq -\beta Q$ | no |
| (i) $Z \leq Q$ | **yes** |
| (j) $p_4, Q^2, Z \geq 0$ | — |

**Proof.** (a) is Equation 40 summed. (b) For $b < 0$ we have $b^3 \geq -n^{-3}$ by (F1), so $b^5 = b^3 b^2 \geq -n^{-3} b^2$; for $b \geq 0$, $b^5 \geq 0$. (c) $b^2 \leq \beta^2$ by (F1), since $\beta \geq 1/n$ for $n \geq 2$.

(d) $Y_R = \sum_i q_i^2 \leq (\max_i q_i) \sum_i q_i \leq MQ$. Equality holds when all $q_i$ agree, in particular at every permutation matrix.

(e) By Equation 40, $f_3(i) \leq \beta q_i$, and $q_i \geq 0$, so $\Gamma_c = \sum_i q_i f_3(i) \leq \beta \sum_i q_i^2 = \beta Y_R \leq \beta MQ$ by (d).

(f) $|\Gamma_a| = |q^T B q'| \leq \|q\| \|B\|_{\mathrm{op}} \|q'\| \leq \sqrt{Y_R}\sqrt{Y_C} \leq MQ$, using (F4) and then (d).

(g) By Cauchy–Schwarz and then (F4),

$$|\Gamma_b| \leq \left( \sum_{i,j} b_{ij}^4 \right)^{1/2} \left\| B B^T B \right\|_F \leq \sqrt{p_4}\, \|B\|_{\mathrm{op}} \left\| B^T B \right\|_F \leq \sqrt{p_4}\sqrt{Z} \leq \beta\sqrt{Q} \cdot \sqrt{Q}, \tag{41}$$

using (c) and (i).

(h) Multiply (a) by $Q \geq 0$ and apply (F3): $Qp_3 \geq -Q^2/n \geq -\frac{n-1}{n} Q$.

(i) is (F4). (j) is clear. ■

## 10.6 Proof of Theorem G

Combining equations 37–39 with Lemma S3 term by term — discarding the nonnegative invariants that carry a plus sign, by (j) — gives lower bounds $\sigma_m(B) \geq -C_m(n)\, Q$ with

$$C_3(n) = \frac{2}{3n}, \quad C_4(n) = \frac{3}{2}\beta, \quad C_5(n) = \frac{24}{5n^3} + \frac{10}{3}\beta + 8\beta^2. \tag{42}$$

For $C_4$: the only negative terms of Equation 38 are $-\frac{3}{4}(Y_R + Y_C)$, bounded by $\frac{3}{2} MQ \leq \frac{3}{2}\beta Q$ using (d) and (F2). For $C_5$: the six terms of Equation 39 contribute $\frac{24}{5} n^{-3}$, $\frac{1}{3}\beta$, $M \leq \beta$, $2\beta$, $4\beta M \leq 4\beta^2$ and $4\beta M \leq 4\beta^2$ respectively, by (b), (h), (f), (g) and (e).

By Equation 35 it therefore suffices that

$$\Phi(n,k) := \frac{4}{t_2} \sum_{m=3}^{k} t_m\, C_m(n) < 1. \tag{43}$$

Using the ratio law Equation 31, $\Phi$ is an explicit rational function of $n$ for each $k$:

$$\Phi(n,3) = \frac{8}{3(n-2)^2}, \tag{44}$$

$$\Phi(n,4) = \frac{16}{3(n-2)^2} + \frac{12n(n-1)}{(n-2)^2(n-3)^2}, \tag{45}$$

$$\Phi(n,5) = \frac{8}{(n-2)^2} + \frac{36n(n-1)}{(n-2)^2(n-3)^2} + \frac{24n^3\,C_5(n)}{(n-2)^2(n-3)^2(n-4)^2}. \tag{46}$$

For $k=2$ the sum in Equation 43 is empty, so $\Phi(n,2)=0$ and the conclusion holds for every $n \geq 2$.

From Equation 44, $\Phi(n,3) < 1$ exactly when $(n-2)^2 > 8/3$, that is $n \geq 4$. For $k=4$ and $k=5$ the same conclusion is reached without any appeal to numerics, as follows. Write each $\Phi(\cdot,k)$ in lowest terms as a ratio of integer polynomials in $n$ with no common content. The denominators are then

$$3(n-2)^2, \quad 3(n-2)^2(n-3)^2, \quad 5(n-2)^2(n-3)^2(n-4)^2 \tag{47}$$

for $k=3,4,5$ respectively, each positive for $n>4$. So $\Phi(n,k)<1$ is equivalent to $P_k(n) > 0$, where $P_k$ denotes the denominator minus the numerator:

$$P_3(n) = 3n^2 - 12n + 4, \tag{48}$$

$$P_4(n) = 3n^4 - 30n^3 + 59n^2 - 48n - 36, \tag{49}$$

$$P_5(n) = 5n^6 - 90n^5 + 445n^4 - 1760n^3 + 620n^2 + 2400n - 3456. \tag{50}$$

Substituting $n = m + n_k$, where $n_k$ is the threshold, makes every coefficient nonnegative with the constant term positive:

$$P_3(m+4) = 3m^2 + 12m + 4, \tag{51}$$

$$P_4(m+8) = 3m^4 + 66m^3 + 491m^2 + 1280m + 284, \tag{52}$$

$$P_5(m+14) = 5m^6 + 330m^5 + 8845m^4 + 121160m^3 + 861620m^2 + 2716720m + 1660864. \tag{53}$$

Since $m \geq 0$ corresponds to $n \geq n_k$, this settles every $n \geq n_k$ at once. The thresholds are sharp for this argument: $P_3(3) = -5$, $P_4(7) = -568$ and $P_5(13) = -306876$, so $\Phi \geq 1$ at $n = n_k - 1$ in each case.

The values at the thresholds are $\Phi(8,4) = 0.8948...$ and $\Phi(14,5) = 0.8094...$, and $\Phi(\cdot,k)$ is decreasing above the threshold, reaching $0.1351...$ at $(k,n) = (4,15)$ and $0.2540...$ at $(k,n) = (5,20)$ and tending to 0 like $n^{-2}$. This proves Theorem G. ■

The binding term throughout is $Y_R$ (with $Y_C$), whose estimate (d) is attained at the permutation matrices. The thresholds in Theorem G therefore cannot be lowered by sharpening (d). They can be lowered by retaining the three nonnegative terms of Equation 38 that the proof discards; at a permutation matrix those terms make $\sigma_4(B)$ positive, whereas Equation 42 allows it to be as negative as $-\frac{3}{2}\beta Q$. §10.11 determines how much that route can give, and shows that it stops short of $\sigma_4 \geq 0$.

### 10.7 The cell $(3,3)$ is a genuine exception

**Proposition S4.** For $n = k = 3$ the conclusion of Theorem G is false.

Lean: `not_stabilityAt_three_three` in `TverbergStability.lean` (commit `1507013`), with the witness stored as `witness33` rather than cited.

**Proof.** Let $P$ be a permutation matrix of order 3 and take $A = \frac{1}{2}(J_3 - P)$, the doubly stochastic matrix that is uniform off a permutation:

$$A = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}. \tag{54}$$

Then $\operatorname{per}(J_3 - P) = 2$, so $\sigma_3(A) = \operatorname{per}(A) = 2/8 = 1/4$, while $\binom{3}{3}^2 \cdot 3!/3^3 = 2/9$. Hence $F = 1/4 - 2/9 = 1/36$. Also $Q = \|A - J_3/3\|_F^2 = 1/2$, so $F/Q = 1/18$. But $c(3,3) = t_2/4 = 1/12 > 1/18$. ■

This is the only cell among those excluded from Theorem G that is known to be a counterexample. For $(4, n)$ with $4 \le n \le 7$ and $(5, n)$ with $5 \le n \le 13$ the inequality is not decided here.

## 10.8 Sharpness

**Proposition S5.** Let $k \ge 3$ and let $c_{\mathrm{opt}}(n,k)$ be the largest constant for which the conclusion of Theorem G holds. Then in the range covered by Theorem G,

$$\frac{1}{4}t_2 \le c_{\mathrm{opt}}(n,k) < \frac{1}{2}t_2. \tag{55}$$

**Proof.** The lower bound is Theorem G with Equation 30. For the upper bound, take $B = J_n/n - P$ for a permutation matrix $P$ and consider $A_s = J_n/n + sB$ for small $s > 0$, which is doubly stochastic for $0 \le s \le 1/(n-1)$. The entries of $B$ are $-\beta$ at the $n$ cells of $P$ and $1/n$ elsewhere, so

$$p_3(B) = -n\beta^3 + \frac{n^2 - n}{n^3} = \frac{n-1}{n^2}(1 - (n-1)^2) < 0 \quad (n \ge 3). \tag{56}$$

By Lemma S1 and equations 36–37,

$$\frac{F(sB)}{\|sB\|_F^2} = \frac{1}{2}t_2 + s\,t_3\,\frac{\frac{2}{3}p_3(B)}{Q_B} + O(s^2), \tag{57}$$

which is strictly less than $\frac{1}{2}t_2$ for all small $s > 0$. ■

So the constant of Theorem G is optimal to within a factor two, and the factor two cannot be closed to a factor one: no constant as large as $\frac{1}{2}t_2$ works for any $k \ge 3$.

One measurement corroborates the picture at $k = 4$, and it is a verified computation, not a theorem: the largest admissible constant, measured against the claimed $c(n,4)$, is 1.88, 1.91, 1.93 at $n = 8, 9, 10$ — rising toward the factor-two ceiling of Proposition S5 and never reaching it.

## 10.9 Verification

Every displayed identity and estimate of this part is checked over the rationals, with no floating-point arithmetic in any decision, by the standalone script `graded_verify_stability.py`; its output is `results/graded_verify_stability.log`. The script imports nothing beyond the standard library. It verifies, in order: Lemma S1 against brute-force subpermanent sums; the core expansions, equations 36–39, against brute-force subpermanent sums; the four facts of Lemma S2, with (F4) tested both directly on random rational vectors and through its consequence $Z \le Q$; each estimate of Lemma S3 in the one direction the proof uses, reporting its slack ratio so the two tightness claims in the table are checkable; the per-layer bounds of Equation 42; the closed forms of equations 44–46, the thresholds and the exceptional sets; the four quoted values of $\Phi$ and its monotonicity above each threshold; the polynomial argument of §10.6, by building $\Phi(\cdot, k)$ as an exact rational function, reducing it to lowest terms as an integer pair, cross-checking it against the arithmetic evaluation, and confirming the sign pattern of $P_k(m + n_k)$, the sharpness values $P_k(n_k - 1) \le 0$, and every denominator and coefficient displayed in §10.6 term by term; Theorem G end to end on doubly stochastic matrices at every covered cell listed in §10.6; Proposition S4 with its witness; Proposition

S5 through the closed form for $p_3(J_n/n - P)$; and the formalisation scope stated in §10.10, against the Lean sources themselves.

That last check is the one exception to the script being self-contained: it reads the Lean files out of the repository at named commits. Run with `--no-lean` it skips them, and records in its log that the scope claims of §10.10 went unchecked in that run; it does not pass them quietly.

The test matrices include the permutation matrices, $(J_n - P)/(n-1)$, the barycentre, random rational convex combinations of permutation matrices, and near-vertex matrices — that is, the configurations at which the estimates of Lemma S3 are attained or nearly attained.

The verifier carries mutation controls, which were run and which fire: four deliberate faults — the constant doubled, a sign flip in Equation 38, each threshold lowered by one so that an excluded cell is claimed as covered, and the estimate of Lemma S3(a) over-tightened — are each rejected by the check responsible for them. With no fault injected those same four checks raise nothing, so each rejection is attributable to its fault. A verifier that never rejects would establish nothing, and `--mutate` reruns the controls alone.

## 10.10 What is machine-checked, and what is not

**What is machine-checked.** Theorem G is kernel-checked at $k = 2$, for every $n \geq 2$, at $k = 3$, for every $n \geq 4$, and at $k = 4$, for every $n \geq 8$ — in each case the whole range the statement covers for that $k$, with no arithmetic gap between the formalised and the written range. The $k = 2, 3$ cells are `stabilityAt_two` and `stabilityAt_three` in `StabilityK3.lean` (commit `1507013`); the $k = 4$ cell is `stabilityAt_four` in `StabilityK4.lean` (commit `944b517`). All three discharge `TverbergStability.StabilityAt k n`: the displayed inequality of Theorem G, with the constant `cVal` equal to $c(n,k)$ and $\sigma_k$ as the subpermanent sum. The $k = 4$ proof follows this part's route step for step: the layer identity at $k = 4$, the core expansion Equation 38 as `sigma_four_centred` (`SigmaFour.lean`, commit `365e44e`), the slice fact (F2) and estimate (d) with the discard of the three nonnegative terms, and the threshold arithmetic consumed as the committed `Phi4_lt_one`, with the bridge `layer_ratio_lt` checking that $\Phi$ rebuilt from the Lean coefficients is exactly Equation 45. Lemma S1, the layer identity that §10.3 runs on, is kernel-checked at **every** $k$, not only at the cells above: `layer_identity` in `LayerIdentity.lean` (commit `27bda3f`), 28 of 28 declarations audited. Its hypotheses are weaker than this part's — it assumes $1 \leq k \leq n$ and $\sum_{ij} B_{ij} = 0$, the single scalar constraint, where Lemma S1 as stated here assumes $A$ doubly stochastic. The doubly stochastic case is the corollary. So the identity underneath the whole argument holds, formally, on a strictly larger set than the theorem needs.

Two limits of the $k = 4$ formalisation must not be over-read, and we state them where the claim is made. First, the threshold 8 is a limit of the argument, not known sharp: the cells $(k = 4, 4 \leq n \leq 7)$ are undecided, and there is no $k = 4$ analogue of `three_threshold_not_slack` — nothing here should be read as implying one. Second, the binding atom is $Y_R$ (with $Y_C$), whose estimate is attained at the permutation matrices, so the threshold cannot move by sharpening anything in the formalised file; it moves only by retaining the three discarded nonnegative terms of Equation 38, the route whose limit §10.11 measures.

Three further claims of this part are kernel-checked alongside those. Two sit in `TverbergStability.lean`: Proposition S4, with its witness stored rather than cited (`witness33`, `not_stabilityAt_three_three`), and the threshold layer of §10.6 at $k = 3, 4, 5$ — the cleared numerators $P_k$ of $1 - \Phi(\cdot, k)$, their shifted forms $P_k(m + n_k)$ with the coefficients displayed there, and $\Phi(n,k) < 1$ above each threshold (`phiPoly3/4/5`, `phiPoly3/4/5_pos`, `Phi3/4/5_lt_one`). The third sits in `StabilityK3.lean`: the sharpness of the $k = 3$ threshold, in the strong form that assuming stability at every $n \geq 3$ is contradictory (`three_threshold_not_slack`).

**What is not.** At $k = 5$ the estimates that feed the budget are not fully formalised, and so neither is Theorem G there: there is no `stabilityAt_five`, and the $(k = 5, n \geq 14)$ cell rests on written proofs plus this part's verifier. Five of the $k = 5$ per-invariant atoms **are** kernel-checked — `StabilityK5Atoms.lean` (commit `f2bdc7f`), whose header lists what is not there — but the expansion Equation 39, the estimates (f) and (g), the fact (F4) and the assembly are not, so the cell is not closed and its grade does not change. Partial atom coverage is not cell coverage, and we do not average the two.

**How that division is checked, not asserted.** The verifier's V12 reads the Lean files above, each at the commit cited for it — `git show`, so the working tree cannot flatter the claim — and confirms each half. On the positive half: that `cVal` **is** $c(n, k)$, compared as exact rationals over $2 \leq k \leq 7$, $2 \leq n \leq 29$ rather than as text; that the hypotheses of `stabilityAt_two`, `stabilityAt_three` and `stabilityAt_four` are the thresholds claimed above; that `sigma_four_centred` is present with both marginal hypotheses in its statement; and that Lean's $P_3, P_4, P_5$, which are an independent derivation, agree at every point with the polynomials this verifier builds for §10.6. It recomputes the two values Lean stores for Proposition S4, $\sigma_3 = 1/4$ and $\|B\|_F^2 = 1/2$, in its own exact arithmetic. On the negative half: that no `stabilityAt_five` exists — with a non-vacuity check that the same search does find the three theorems that are there, since an absence found by a broken search is not an absence.

On Lemma S1 it checks that Lean's coefficient `tVal` is this part's $t_m$, again as exact rationals and not as text, and that `layer_identity` carries every ingredient of the displayed identity while its hypotheses contain no double stochasticity — with the matching non-vacuity check that the token **is** findable, since it occurs in `StabilityK3.lean`. It runs the orphan diff on the four files that carry audit blocks, empty in both directions: 33 of 33 declarations in `StabilityK3.lean` carry `#print axioms` lines, as do 14 of 14 declarations in `StabilityK4.lean` and 46 of 46 declarations in `SigmaFour.lean`, with 28 of 28 in `LayerIdentity.lean`; no line in any of the four lacks a declaration, and no source contains `sorry` or `native_decide` outside its comments. All four counts are parsed back out of this section, so a number stated here and a number measured there cannot drift apart.

One limit of that check is worth stating. The axiom sets themselves — that each of the 121 audited declarations reports a subset of `propext`, `Classical.choice`, `Quot.sound`, with no `sorryAx` — come from elaborating the files, which only Lean can do and which this verifier does not repeat. V12 checks that each audit block is complete and that the sources are free of `sorry`; the conformance of what those blocks print is the Lean build's evidence, not this script's.

## 10.11 Addendum: the limit of the discarded-terms route

§10.6 discards the three nonnegative invariants of Equation 38 and notes that retaining them would lower the thresholds. This section determines by how much. The answer is bounded, and the natural target fails.

**Proposition S6.** $\sigma_4$ restricted to the centred slice is indefinite. At $n = 4$, the doubly stochastic matrix

$$A = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \tag{58}$$

has $B = A - J_4/4 = \frac{1}{4}M_0$ with

$$M_0 = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & 1 & 1 & -1 \end{pmatrix}, \tag{59}$$

and $\mathrm{per}(M_0) = -8$, so $\sigma_4(B) = -1/32$ while $Q = 1$.

Rows three and four of $M_0$ are the negatives of rows one and two, so $M_0$ has rank two; all its line sums vanish. Two independent searches over $\Omega_4$ — random sampling of rational convex combinations of permutation matrices, and local descent by $2 \times 2$ cycle moves — both bottom out at $\sigma_4/Q = -1/32$ exactly, at matrices of this shape.

So the hoped-for $\sigma_4 \geq 0$ is false, and the route can at best replace $C_4$ of Equation 42 by a constant $\varepsilon$ with $\varepsilon \geq 1/32$. Doing so would give thresholds $(k = 3, n \geq 4)$, $(k = 4, n \geq 5)$, $(k = 5, n \geq 12)$, against Theorem G's $(4, 8, 14)$; the same thresholds result for every $\varepsilon \leq 1/16$, so the exact constant does not matter to the outcome. The available gain is therefore three steps at $k = 4$, two at $k = 5$, and none at $k = 3$. Proving any such $\varepsilon$ is a separate problem: the inequality is tight along a family (below), false at $\varepsilon = 0$, and the elementary term-by-term bounds of §10.5 give only $\varepsilon \approx 9/8$. This unproved $\varepsilon$ is exactly the hypothesis of the conditional statement in §11.1, and it is recorded there as a parked research problem.

The equality family of §10.5 has an explanation, which also locates the negativity.

**Proposition S7.** For centred $u, v$ (that is, $\sum_i u_i = \sum_j v_j = 0$) and $B = c\,uv^T$, which is automatically doubly centred,

$$\sigma_4(B) = \frac{3}{2} c^4 D(u) D(v), \quad D(x) = \sum_i x_i^4 - \frac{1}{2}\left(\sum_i x_i^2\right)^2. \tag{60}$$

The $2 \times 2$ cycle directions are precisely the locus $D = 0$: for $u = e_i - e_{i'}$ one has $\sum u^4 = 2$ and $\left(\sum u^2\right)^2 = 4$, so $D(u) = 0$. That is why $\sigma_4$ vanishes identically along them rather than by accident. The zero locus is larger than the two-entry vectors — $D(2, -1, -1, 0) = 0$ as well.

$D$ takes both signs: $D(1, 1, -1, -1) = -4$ and $D(3, -1, -1, -1) = 12$. Choosing $D(u) > 0$ against $D(v) < 0$ makes $\sigma_4$ negative on a rank-one direction inside $\Omega_4$: with $u = (3, -1, -1, -1)$, $v = (1, 1, -1, -1)$ and $c = 1/12$, the entries of $J_4/4 + B$ are nonnegative and $\sigma_4(B)/Q = -1/96$. An exhaustive search over rank-one directions with small integer $u, v$ inside $\Omega_4$ bottoms out at exactly that value, three times smaller in magnitude than the rank-two minimum $-1/32$. So negativity begins at rank one but the extremum is not attained there, and a proof for rank-one directions alone would not settle the question.

**On novelty.** Tverberg's inequality and Friedland's proof give strictness but no rate. We are not aware of a previous quantitative form, and Theorem G appears to be the first stability form of the Tverberg–Friedland theorem. The claim of machine-checked status divides with the cells, as §10.10 sets out: at $k = 2$, $k = 3$ and $k = 4$ the theorem is kernel-checked, and we are not aware of an earlier kernel-checked stability form of Tverberg–Friedland at any $k$; at $k = 5$ the support layer is written proofs plus the exact verifier, and no machine-checked status is claimed for that cell. The narrower scoping is deliberate: quantitative stability bounds for other permanent inequalities do exist, and the claim above is about Tverberg–Friedland specifically.

# 11 The case $k = 4$: confinement, splitting, and the collar

This part proves Theorem H and establishes the insensitivity computation stated beside it. The argument is local. Confinement (Theorem E) traps any violator in a collar around the Birkhoff polytope; a collar matrix splits orthogonally into a line-sum block and a doubly centred block; the two blocks are governed by separate results, and the five cross terms between them have exact reductions, all displayed below. The computation is the second deliverable: at every sampled admissible value of the collar bound on row squared norms — the one constant in the argument that is not settled — the threshold stays at $n \geq 10$, and a mechanical audit pins down where that constant can act at all.

## 11.1 Statement, scope, and the remaining gap

In the centred coordinates of §4, with $F(A) = (2 - \gamma(n, 4)) - \Phi_4(A)$, Theorem H asserts $F(A) > 0$ for all $A \in K_n$ other than $A = J_n/n$, where $F = 0$, for every $n \geq 10$.

**Not formalised.** Theorem H is not machine-checked, and neither is the insensitivity computation. Their support layer is written proofs together with the exact verifier of §11.9, which recomputes every displayed quantity of this part over the rationals.

**What is formalised is a neighbouring statement, on the slice.** The stability form of the Tverberg–Friedland theorem is kernel-checked at $k = 2$, $k = 3$ **and** $k = 4$ on the doubly stochastic slice itself (§10.10; the $k = 4$ cell is `stabilityAt_four`, `StabilityK4.lean`, commit `944b517`). This part consumes that result through the centred-block input at $k = 4$ — so the slice form of the input is now kernel-checked — but everything the collar costs on top of the slice is not: the collar facts of §11.6, the cross-term reductions, the merge and the budget are supported by written proofs plus the verifier only. No claim of machine-checked status attaches to Theorem H.

**The remaining honest gap is $n = 8$ and $n = 9$.** Theorem H begins at $n = 10$, and the anchor certificates of §12 settle $(5, 4)$, $(6, 4)$ and $(7, 4)$, so $k = 4$ is not settled at exactly $n = 8, 9$ by what this paper carries. Two routes close the gap: fixed-$n$ certificates at the two open cells, whose state is §12; or the certificate-family solve in $\mathbb{Q}(n)$, which needs no anchors (§13). Under the hypothesis of the conditional statement below the line would be fully closed: the conditional route reaches $n \geq 8$, and its only outstanding cell, $(7, 4)$, is settled by anchor certificate.

**Conditional statement (the hypothesis is part of the statement; this is not a result of this paper).** Suppose that on the doubly centred slice $\sigma_4(z) \geq -\varepsilon \|z\|_F^2$ for some $\varepsilon \leq 1/16$. Then for $k = 4$ and every $n \geq 8$, $F(A) > 0$ for all $A \in K_n$ other than $J_n/n$.

The hypothesis is **not proved**. It is known to be false at $\varepsilon = 0$, and the measured obstruction is $\varepsilon \geq 1/32$, with the explicit rank-two witness of Proposition S6. It is recorded as a parked research problem, and this box should be read as a conditional statement and not as a theorem.

**Instruments used and not re-derived.** The line block is `pincer_line.py`: `F_line` and the $(S4)$ closed form for $\sigma_d$ of a rank-two line matrix, `line_margin` (margins computed, not tabulated), `u_max`, `lam_line`. The centred block is `pincer_onesided.py`: `deficit_centred`. The decomposition, the identity $\sigma_2\big(z^{(a,b)}\big) = 2z_{ab}^2 - q_a - q'_b + Q/2$, and the $k = 3$ instance of the assembly are `pincer_assembly_k3.py`. The collar bound on row squared norms is the committed cap of `graded_lemmaB.py`, and the operator-norm degradation off the slice is the collar form (K4) of §11.6, the analogue of the slice fact (F4).

## 11.2 The decomposition, stated once

Let $A \in K_n$ and $B = A - J_n/n$ with row sums $R$ and column sums $C$. Set

$$x_i = R_i/n, \quad y_j = C_j/n, \quad L_{ij} = x_i + y_j, \quad z = B - L, \tag{61}$$

so $\sum_i x_i = \sum_j y_j = 0$, $z$ is doubly centred, and $\|L\|_F^2 = n(\mu + \nu)$ with $\mu = \|x\|^2$, $\nu = \|y\|^2$. As in §10.1, but now for the centred block $z$, write $Q = \|z\|_F^2$, $q_i = \sum_j z_{ij}^2$, $q'_j = \sum_i z_{ij}^2$, $N = zz^T$, $N' = z^T z$, and $f_3(a) = \sum_b z_{ab}^3$, $g_3(b) = \sum_a z_{ab}^3$.

Confinement bounds the line block only:

$$\mu + \nu \le u_{\max}(n, k) = \frac{(n-1)\, k!}{n^{k+1}}. \tag{62}$$

### 11.3 The layer identity, and what couples

Theorem C's identity Equation 7 gives $F = \sum_{d=1}^{k} [t_d\, \sigma_d(B) - s_d(e_d(R) + e_d(C))]$, with the same coefficients $s_d, t_d$ that §10.1 writes in the layer index $m$. Two facts make the split usable.

**The $e_d$ half never couples.** $z$ has vanishing line sums, so the row and column sums of $L + z$ are those of $L$ alone.

**Expansion by the number of $z$ factors.** For any $X, Y$ and any $d$,

$$\sigma_d(X + Y) = \sum_{j=0}^{d} \sum_{|S| \,=\, |T| \,= j} \operatorname{per}(X[S|T])\, \sigma_{d-j}\big(Y^{(S,T)}\big), \tag{63}$$

$Y^{(S,T)}$ being $Y$ with rows $S$ and columns $T$ deleted. Taking $X = z$, $Y = L$ and collecting,

$$F(L + z) = F_{\text{line}}(x, y) + F_{\text{centred}}(z) + \sum_d t_d \cdot (\text{cross parts of } \ \sigma_d). \tag{64}$$

At $d = 2$ the cross part vanishes **identically**, not merely to leading order.

### 11.4 The five cross-term reductions

All five are exact and are verified against brute force in §11.9.

At $d = 3$:

$$X_1 = (n-2)^2\, x^T z\, y, \quad X_2 = (2 - n)\, \Xi, \quad \Xi := \sum_i x_i q_i + \sum_j y_j q'_j. \tag{65}$$

At $d = 4$:

$$Y_1 = -(n-2)(n-3)^2 \big[\, x^T z\, (y \circ y) + (x \circ x)^T z\, y \,\big], \tag{66}$$

$$Y_2 = (n-2)(n-3) \left[ -\sum_{s_1 < s_2} N_{s_1 s_2} e_2(x_I) - \sum_{t_1 < t_2} N'_{t_1 t_2} e_2(y_J) \right] + 2(n-3)^2 \sum_{i,j} z_{ij}^2 x_i y_j, \tag{67}$$

$$Y_3 = -(n-3)[\, 2A_1 - A_2 + 2A_3 - A_4 \,], \quad A_1 = \sum_a x_a f_3(a), \quad A_2 = x^T z\, q', \quad A_3 = \sum_b y_b g_3(b), \quad A_4 = q^T z\, y, \tag{68}$$

where $\circ$ is the entrywise product, and $I = [n] \setminus S$, $J = [n] \setminus T$ for the deleted index pairs $S = \{s_1, s_2\}$, $T = \{t_1, t_2\}$.

Three mechanisms produce them, and each is the same principle: anything separable in the deleted indices is annihilated by $\sum_a z_{ab} = \sum_b z_{ab} = 0$.

**(a) $Y_1$: the restricted $e_1$ is not zero.** On $I = [n] \setminus \{a\}$ and $J = [n] \setminus \{b\}$ one has $e_1(x_I) = -x_a$, $e_1(y_J) = -y_b$ and $e_2(y_J) = e_2(y) + y_b^2$. In the $(S4)$ closed form for $\sigma_3$ the $t = 0$ and $t = 3$ terms depend on $b$ alone and on $a$ alone and die; the $t = 1, 2$ terms leave $x_a y_b^2$ and $x_a^2 y_b$ with coefficient $-(n-2)(n-3)^2$.

**(b) $Y_2$: a vanishing lift.** Its $e_2$ pieces depend on $S$ alone and $T$ alone, and $\sum_T \operatorname{per}(z[S|T]) = -N_{s_1 s_2}$, $\sum_S \operatorname{per}(z[S|T]) = -N'_{t_1 t_2}$. For the $x_S y_T$ piece, lift the restricted sum to the unrestricted one over all four indices: every expanded piece carries a lone $\sum_s z_{st}$ or $\sum_t z_{st}$, so the lift is **identically zero** and the restricted sum is a pure diagonal correction, equal to $2\sum_{ij} z_{ij}^2 x_i y_j$. An absolute-value bound discards precisely that cancellation and overshoots by four orders of magnitude.

**(c) $Y_3$: subpermanents through a row.** $\sigma_1\big(L^{(S,T)}\big) = -(n-3)(x_S + y_T)$, and $\sum_{S,T} \operatorname{per}(z[S|T])\, x_S = \sum_a x_a \Psi_a$ with $\Psi_a = \sum_b z_{ab}\sigma_2\big(z^{(a,b)}\big) = 2f_3(a) - (zq')_a$.

## 11.5 The merge, and why $k = 4$ inherits the clean form

On the collar $A \geq 0$ gives the **per-entry** bound

$$z_{ij} \geq -\Big(\frac{1}{n} + x_i + y_j\Big), \tag{69}$$

not $z_{ij} \geq -1/n$. Cubing and summing,

$$\sum_{ij} z_{ij}^3 \geq -\frac{1}{n}Q - \Xi, \tag{70}$$

and the perturbation is **exactly** the invariant of the cross term $X_2$. The two effects are therefore one quantity, to be charged once — but **at its full summed coefficient**: the $m = 3$ layer contributes $\frac{2}{3}\Xi$ and the cross term $(n-2)\Xi$, so the coefficient is $(3n-4)/3$, not $(n-2)$. Counting once deletes the double count, not a coefficient.

At $k = 4$ the layers are $m = 2, 3, 4$, and the only odd-power one-sided step is at $m = 3$: the centred core of $\sigma_4$ is Equation 38 — $\frac{3}{2}p_4 + \frac{1}{8}Q^2 + \frac{1}{4}Z - \frac{3}{4}(Y_R + Y_C)$, in the invariants of §10.1 now taken of $z$ — every term of even degree. So no second perturbation can arise and the merge keeps its single-coefficient form. This is special to $k \leq 4$: at $m = 5$ the perturbation enters as $\big(1/n + x_i + y_j\big)^3$, which is four cross invariants rather than one.

## 11.6 The collar facts

Write $\rho^2 = (n-1)k!/n^{k-1}$ for the confinement radius on line sums, and $\beta_c = 1 + \rho - 1/n$. The four facts below are the collar forms of (F1)–(F4) of Lemma S2, in the same order, with the degradation the collar costs; the verifier's log names them G1–G4.

**(K1)** $-1/n \leq b_{ij} \leq \beta_c$. The lower side needs only $A \geq 0$ and so does **not** degrade off the slice; that asymmetry is what makes the $m = 3$ step cheap. **(K2)** $\max_i q_i(B) \leq (1+\rho)^2 - 1/n + 2\rho/n$, from $\sum_j A_{ij}^2 \leq \big(\max_j A_{ij}\big)\Big(\sum_j A_{ij}\Big) \leq (1+\rho)^2$. **(K3)** $Q \leq n - 1 + \rho^2$, since $Q = \sum_{ij} A_{ij}^2 - 1$ and $\sum_{ij} A_{ij}^2 \leq \sum_i (1 + R_i)^2 = n + p_2(R)$. **(K4)** $\|B\|_{\mathrm{op}} \leq \sqrt{(1+\rho)^2 + \rho^2/n}$: a nonnegative matrix with all line sums at most $1 + \rho$ has operator norm at most $1 + \rho$, and $Bu = R/\sqrt{n}$ for $u = \mathbf{1}/\sqrt{n}$. Birkhoff is lost off the slice; the bound is not. Consequently $\|z\|_{\mathrm{op}} \leq \|B\|_{\mathrm{op}} + \sqrt{2n\, u_{\max}}$.

**The one unsettled constant.** The slice bound $q_i(z) \leq 1 - 1/n$ requires row sums equal to 1 **and** nonnegativity; on the collar $J_n/n + z$ has row sums 1 but can have negative entries, so the bound does not transfer, and it is refuted — a permutation with one reweighted row violates it by a factor 1.63 at $n = 10$ and 1.58 at $n = 11$, and violations occur from $(k, n) = (3, 4)$ upward at perturbations as small as $t = 1/60$. We therefore write $q_i(z) \leq c\,(1 - 1/n)$ and carry $c$ symbolically. §11.8 shows the threshold does not depend on it.

## 11.7 The budget

Layer two is the budget: on the centred block $t_2\sigma_2(z) = \frac{1}{2}t_2 Q$, and on the line block $F_{\text{line}} \geq \frac{1}{2}\lambda_{\text{line}}(\mu + \nu)$. Every other contribution is bounded below and charged to whichever budget it scales with — a term carrying $(\mu + \nu)$ to the line budget, one carrying $Q$ to the centred budget. $X_1$ and $Y_1, Y_2$

carry $(\mu+\nu)$ or $(\mu+\nu)^{3/2}$ and go to the line side; $\Xi$ and $Y_3$ carry $Q$ or $\sqrt{\mu+\nu}\sqrt{MQ}$, with $M = \max\bigl(\max_i q_i, \max_j q'_j\bigr)$ as before, and go to the centred side, $Y_3$ split between the two by the arithmetic–geometric mean. The resulting consumption, as a fraction of each budget, is what §11.9 recomputes line by line; both totals below 1 is the conclusion.

## 11.8 Insensitivity to the unsettled constant

This is a computation with an audited structure, graded as exactly that in its statement in the introduction; what follows is the record. Only three budget lines involve $c$: the $\sigma_4$ core line is **linear** in it, and the two $Y_3$ lines ride on $\sqrt{c}$. Everything else — the $m = 3$ core, $\Xi$, the line tail, $X_1$, $Y_1$, $Y_2$ — is independent of it. That decomposition is not read off the derivation alone: it is asserted mechanically, every budget line recomputed at $c$ and $2c$, at $n = 10$ and $n = 16$, with the run aborting on any change — so a line that silently acquired a $c$-dependence would kill the computation rather than skew it. At $n = 10$ the linear share of the centred column is 0.379 and the total exposure 0.488; by $n = 20$ these are 0.311 and 0.341. So about half the column is insulated, and the linear part carries the already-small factor $t_4/t_2$.

| $c$ | honest threshold |
|---|---|
| 1.00 (slice value; refuted on the collar) | 9 |
| 1.25 | 9 |
| 1.50 | 9 |
| 1.58 (smallest constructed violation, at $n = 11$) | 10 |
| 1.63 (largest constructed violation, at $n = 10$) | 10 |
| 2.00 | 10 |
| 2.34 (collar cap, low) | 10 |
| 2.53 (collar cap, high) | 10 |
| 3.00 | 11 |
| 4.00 | 11 |

Table 7: The threshold of Theorem H at the ten sampled values of the collar constant $c$. The admissible band is $1.58 \le c \le 2.53$, and the threshold is 10 at every sampled value in it. Every row is parsed out of this document and recomputed by the verifier of §11.9.

The admissible band is $1.58 \le c \le 2.53$: below it the constructed violation forbids, above it the proved cap forbids. The threshold is 10 at every sampled value in the band, and the audited structure above names the only three lines through which an unsampled $c$ could act; no claim is made for unsampled values. What the computation does establish is direction: sharpening the constant cannot improve Theorem H at any sampled value, and no sampled weakening within the cap damages it. On the route of the conditional statement the picture differs, because there the $\sigma_4$ core line is replaced by $\varepsilon$ and is $c$-free while the $Y_3$ lines are not: the cap is worth one step, $n \ge 9$ becoming $n \ge 8$.

## 11.9 Verification

`graded_verify_k4.py` recomputes every displayed quantity of this part over $\mathbb{Q}$ with no floating-point arithmetic in any decision, and **reads the numbers out of this document** — the sensitivity table of §11.8 is parsed from the typst source of this paper, not restated in the script — so that displayed and checked cannot drift apart. It checks: the expansion Equation 63 at $d = 2, 3, 4$ against brute force, including that the $d = 2$ cross part is exactly zero; the end-to-end identity Equation 64 with every $t_d$ in place, against $F$ computed from the 1992 functional; all five reductions of §11.4 against brute force; the per-entry bound and the merge of §11.5, on configurations of **both** signs of $\sum z^3$; the four collar facts of §11.6; every budget line of §11.7 recomputed from the layer identity; and the sensitivity table row by row against the parsed document. Its output is `results/graded_verify_k4.log`.

Mutation controls, each with a **separating** witness — one on which the injected fault actually changes the value, asserted in the same line:

| **control** | **fault** | **caught by** |
|---|---|---|
| M1 | a $t_d$ factor dropped from a cross term | end-to-end identity |
| M2 | the merge coefficient $(3n-4)/3$ replaced by $(n-2)$ | budget-line check |
| M3 | the $d = 2$ cross part assumed nonzero | expansion check |
| M4 | $c$ rescaled in a $c$-independent line | sensitivity audit |

Table 8: The four mutation controls of `graded_verify_k4.py`.

Two of these encode errors made and corrected while the argument was being assembled: a dropped $t_4$, and the merge coefficient. They are controls because they happened.

## 12 The anchor cells: $k = 4$, $5 \leq n \leq 9$

Theorem H begins at $n = 10$. The five cells below it are each a fixed-$n$ question, and each is recorded here at the grade it has actually reached — no cell inherits the grade of a neighbour. **Anchor grade** — the $(5, 4)$ standard — means an exact rational certificate accepted in full by the six checks of the standalone verifier of §7.1: the bound, $\sigma_k$ by two structurally different algorithms, the polynomial identity by **full** coefficient comparison, positive definiteness of the assembled Gram matrices by exact $LDL^T$ over $\mathbb{Q}$, equality only at $J_n/n$, and mutation tests rejected. Anything less is stated as what it is.

| cell | status | support |
|---|---|---|
| $(5,4)$ | settled: $\Phi_4 \le 1226/625$ on $K_5$, equality only at $J_5/5$ | anchor grade: the stored certificate `subdittert_n5k4d2_certificate.json` of §7.1 — 26 exact $LDL^T$ factorisations of size 350, full coefficient comparison, mutation tests rejected; two independent full runs logged |
| $(6,4)$ | settled: $\Phi_4 \le 107/54$ on $K_6$, equality only at $J_6/6$ | anchor grade, on the stored exact witness `n6_H2_201.json`: all six checks of the $(5,4)$ standard — the full-monomial identity over all 40,386 coefficients; both assembled Gram matrices (cone size 702) positive definite over $\mathbb{Q}$, least pivots $4.083407 \times 10^{-5}$ ($\sigma_0$) and $1.070308 \times 10^{-5}$ ($\sigma_{11}$); conjugacy proved exactly, all 36 transporters inducing basis bijections with the corner Gram invariant under the stabiliser; equality only at $J_6/6$; mutation tests rejected |
| $(7,4)$ | settled: $\Phi_4 \le 4778/2401$ on $K_7$, equality only at $J_7/7$ | anchor grade, on the stored exact witness `n7_H2_201.json`: all six checks of the $(5,4)$ standard — the full-monomial identity over all 156,555 coefficients; $\sigma_k$ by two structurally different algorithms; the bound; equality only at $J_7/7$; mutation tests rejected; conjugacy proved exactly, all 49 transporters inducing basis bijections; and check [4], both assembled Gram matrices positive definite over $\mathbb{Q}$, by the congruence route stated below, every link of which is a stored artefact |
| $(8,4)$ | not settled; one check short of anchor grade | on the stored exact witness `n8_H2_201.json`, five of the six checks of the $(5,4)$ standard hold: the full-monomial identity over all 496,448 coefficients, computed in two parts that sum exactly; $\sigma_k$ by two structurally different algorithms; the bound 1021/512; equality only at $J_8/8$; mutation tests rejected — and the conjugacy is proved exactly, all 64 transporters inducing basis bijections. Positivity holds block-wise: all 21 canonical blocks positive definite over $\mathbb{Q}$, least pivot $1.960 \times 10^{-5}$. What has not run is the assembled $2144 \times 2144$ factorisation — check [4] of the standard — and block definiteness is not assembled definiteness, so the cell is not settled |
| $(9,4)$ | not settled | no witness exists: the exact solve exceeded available memory at this size. Nothing is stored, and nothing is claimed |

Table 9: The five cells between Theorem A's line and Theorem H's range, each at its own grade, as of 30 July 2026.

**How $(7,4)$ closed check [4].** Both assembled Grams are positive definite over $\mathbb{Q}$ by congruence rather than by a single direct factorisation, and each link of the route is a stored artefact. The 21 isotypic components are **exactly** $H$-orthogonal over $\mathbb{Q}$ and their translates span — 1,368,988 inner products checked, every one exactly zero (`anchor_wtest_n67_part1.log`). On each component the congruence block is the Kronecker product of two smaller matrices, $h_b$ and $C_b$, and all 21 of each are positive definite under exact $LDL^T$ (`anchor_cb_measure_n7.log`; the $h_b$ independently in `verify_H2_n7.log`); the Kronecker argument then makes each component block positive definite, orthogonality-with-spanning assembles the components, and the conjugacy (A1)/(A2) extends the verdict to all 49 multiplier Grams (`anchor_check3_n7.log`). The two component computations use different bases of the same components, so their join is itself checked — every component dimension $d \cdot e$ against the spanned dimension, all 21 components, total $2548 = 2 \times 1274$ (`anchor_join_n567.log`), with four rejection controls that fire (`anchor_join_controls.log`). That join check is a consistency check between stored artefacts, not an independent verification: the mathematics is in the links. $\sigma_0$ additionally admits a direct assembled factorisation, positive definite over $\mathbb{Q}$ with least pivot $1.637015 \times 10^{-5}$; the full factorisation record, including $\sigma_{11}$'s, is archived when the run completes, and the congruence route does not depend on it.

With the cells as they stand, the unconditional state of the line $k = 4$ is: settled for $n = 5, 6, 7$ (anchor grade) and for every $n \geq 10$ (Theorem H, written-proof plus exact-verifier grade); not settled at $n = 8$ and $n = 9$. The line becomes settled for every $n \geq 5$ exactly when the two open cells of Table 9 reach at least anchor grade, and not before. None of the theorems of this paper depends on any cell of the table.

# 13 Toward a certificate family at $k = 4$ and $k = 5$

**Support layer: exact rational computation with stored witnesses. Nothing in this section is a theorem of this paper, and nothing in it is Lean-checked.**

The line $k = 4$ now carries Theorem H, but Theorem H is not a certificate: the local route bounds the deficit below by budgeted charges rather than exhibiting an identity, and it neither reaches the open cells $n = 7, 8, 9$ of §12 nor offers anything to formalise by the pattern of Theorem A. A certificate family in $\mathbb{Q}(n)$ at $k = 4$ would do all three at once — supersede the local route, close the open cells in one stroke, and inherit the $k = 3$ formalisation pattern. This section records the decided groundwork toward one.

The certificate programme of §5 has a shape that depends on $k$ only through $e = \lceil (k-1)/2 \rceil$, because a common Gram basis of degree $e$ gives $\deg(b_p \sigma_p) = 2e + 1$, which must reach $\deg F = k$. Mixed degrees are ruled out for even $k$: if $\sigma_0$ carried basis degree $k/2$ and the multipliers $k/2 - 1$, the degree-$k$ part of the identity would make the top form $F_k$ a sum of squares on the hyperplane, and Equation 7 shows $F_k(b) = s_k[s_k \sigma_k(b) - e_k(R) - e_k(C)]$, which is negative at explicit doubly centred integer matrices — the smallest is a $4 \times 4$ with $\mathrm{per}(b) = -854$. So $e(k) = \lceil (k-1)/2 \rceil$ and the programme's counts are constant on the bands $\{2, 3\}$, $\{4, 5\}$, $\{6, 7\}$, and so on: within a band the constraint matrix is literally the same matrix and only the right-hand side moves. Band one is $\{2, 3\}$, which Theorems A and D close.

The same computation bounds the method's reach. The binding block's multiplicity is $2, 16, 93$ at $e = 1, 2, 3$, and the programme grows with it: 19 unknowns at $e = 1$ and 440 at $e = 2$. The route therefore reaches every **fixed** $k$ at all large $n$ and cannot reach $k = n$, where the stabilisation threshold and the parameter collide.

For band two the current state is a decided sweep rather than a certificate. At $k = 4$ the pinning programme was decided over $\mathbb{Q}$ at $n = 5, 6, 7$, with no verdict resting on a solver's reported sign: infeasibility is witnessed by a rational Farkas multiplier, positive semidefinite by construction, whose value is constant and non-positive on the affine set; feasibility by a rational point with every canonical block positive definite under exact $LDL^T$. The full pinning is infeasible at all three dimensions, as are twelve of the thirteen single-block omissions. The one programme that survives — 201 pins, omitting the $16 \times 16$ block of $\sigma_{11}$ — has an exact strictly feasible rational point at every dimension tested, $n = 5, 6, 7, 8$, with least stored pivots $5.11 \times 10^{-5}$, $4.94 \times 10^{-5}$, $4.84 \times 10^{-5}$ and $1.96 \times 10^{-5}$. The cases $n \geq 6$ had resisted every floating-point instrument, and the failure has an exact explanation: three integer relations among the seventy diagonal constraints leave the working coordinates roughly twelve digits below the constant column's scale, so each solver ran beneath its own cancellation floor. A change of variables taking the constraint values themselves as the unknowns — validated first against the known $n = 5$ verdict as a positive control — produced the points. Every verdict has its witness stored as exact rationals and re-checked by a standalone verifier that re-derives the constraint system and the pin rows from the problem definition, carries its own elimination and its own $LDL^T$, and confirms that every canonical basis has full column rank — without which block definiteness would not imply definiteness of the whole and the test would be vacuous. Mutation controls were run and failed as they should.

There is no certificate family in $n$ for band two, so there is nothing yet to formalise.

# 14 Assessment

**What is proved, at what grade.** Theorem A, for every $n \geq 4$, by an exact rational certificate whose coefficients are rational functions of $n$, with definiteness decided by Sturm sequences over $\mathbb{Q}$ and the identity verified independently at six dimensions — machine-checked end to end. Theorems C to F, in Lean, uniformly in $n$ and $k$. Theorem G, kernel-checked at $k = 2$, 3 and 4, over the whole stated range for each of those $k$, and carried at $k = 5$ by written proofs plus the exact verifier of §10.9. Theorem H, by written proofs plus the exact verifier of §11.9, not formalised; the insensitivity computation beside it is graded in its own statement. No floating-point number enters any of these decisions: the numerics find candidates, and every accepted statement is re-established over $\mathbb{Q}$ or over $\mathbb{F}_p$.

**What kind of proofs these are.** Theorem A is a computer proof, and a small one by the standards of the genre — nineteen rational functions of $n$, of degree at most 12. It is checkable in minutes at any fixed $n$. It conveys very little insight into **why** the bound holds; §6.3 is the closest thing to an explanation we can offer, and it is an explanation of the certificate's geometry rather than of the inequality. Theorems G and H are the opposite kind of object: ordinary estimates assembled by hand, whose computer content is verification rather than construction — the exact verifiers recompute what the written proofs display, and in two places parse this document itself so that displayed and checked cannot drift; the insensitivity computation is the same species with a smaller claim. Theorems C to F are ordinary mathematics that happens to be machine-checked.

**What is machine-checked, precisely.** Theorem A itself in all three parts, and with it every step between the 1992 statement and them: the definitions, the tightness of the functional at $J_n/n$ for every $k \leq n$, the rank-one identity that pins the definition of $\sigma_k$, the closed form for $\sigma_3$ and its reduction of the objective to an explicit polynomial, the ten positivity facts for all real $n \geq 4$, positive semidefiniteness of both Gram families and positive **definiteness** of the $\sigma_0$ Gram, the Positivstellensatz identity, the deduction from certificate to bound, the equality case — twice, by independent routes — and the stability bound with its explicit constant. Beyond Theorem A: the universal decomposition at every $(n, k)$, the case $k = 2$ at every $n$, the confinement at every $2 \leq k \leq n$, Newton's and Maclaurin's inequalities, and — for Theorem G — the cells ($k = 2$, every $n \geq 2$), ($k = 3$, every $n \geq 4$) and ($k = 4$, every $n \geq 8$), the layer identity at every $k$ under a weaker hypothesis, the $\sigma_4$ core expansion at every centred matrix, the threshold polynomials at $k = 3, 4, 5$, the $(3, 3)$ counterexample with its stored witness, and the non-slackness of the $k = 3$ threshold — subject to the two stated limits of the $k = 4$ formalisation, which §10.10 carries. §9.1 states the standard and pins it to a commit; §10.10 does the same for the stability files at their own commits. What the machine does not check is that the Lean definitions say what the 1992 paper says; §9.2 item 2 is the argument that they do. We state the division explicitly because "machine-checked" is otherwise an ambiguous claim.

**What is not claimed.** At $k = 4$: nothing beyond Theorem H's range $n \geq 10$, the settled cells $(5, 4)$, $(6, 4)$ and $(7, 4)$, and the per-cell records of §12, each at its stated grade — the cells $n = 8$ and $n = 9$ are open here. At $k = 5$: nothing beyond the stability cell ($k = 5, n \geq 14$) of Theorem G, at written-proof plus exact-verifier grade; the Cheon–Hwang inequality itself is not claimed at any $(5, n)$ except the anchor $(5, 4)$ of §7.1, which concerns $k = 4$. No statement about general $k$ beyond Theorems C and E. Nothing refereed. The constant $\theta_2(n)$ of Equation 5 is not claimed optimal (§9.4), and the constant $c(n, k)$ of Theorem G is optimal only to within the factor two of §10.8. No novelty is claimed for the identity of Theorem C, for the statement of Theorem D, for Theorem E, or for Theorem F. We have not read Cheon and Wanless 2007 [7] and do not rely on it.

**On the priority claims.** §2 records public dated claims on the $k = n$ endpoint that appeared within a single week of this work, three of them in public repositories [11] with no corresponding preprint, and one assembly among them claiming Dittert in full. To our knowledge Theorem A is the first resolved case of the Cheon–Hwang conjecture with $2 < k < n$, and Theorem G is the first stability form of the Tverberg–Friedland theorem (§10.11 states that claim's scope). Both claims cover public code repositories as well as the indexed literature; both are priority claims, not claims of difficulty; and both are perishable on a line that has moved this fast — they should be re-checked before this paper is submitted anywhere. Theorem H extends the resolved range to a second line at a lower grade, and its priority claim is subordinate to its grade: written proofs plus an exact verifier, not machine-checked, not refereed.

**Where the methods stop.** $\deg F = k$, so the Gram basis degree must grow with $k$, and the degree-counting obstruction that blocks the $k = n$ line at even dimensions reappears in $k$; §13 quantifies it. The line $k = 3$ is tractable exactly because its degree budget is fixed while the dimension grows. The local route of §11 stops where its merge does: at $m = 5$ the one-sided step splinters into four cross invariants (§11.5), so $k = 5$ needs new ideas on the centred block, not more of the same. The natural next targets are the open cells of §12 by the fixed-dimension method, and — a genuinely different question — whether $k = 4$ admits the same uniform certificate treatment as $k = 3$ (§13).

# 15 Data availability

The certificates as exact rational data, the objective builders, the block-diagonalisation, the Sturm decision, the independent verifiers with their positive and negative controls, and the Lean development are supplied as a reproduction kit accompanying this paper. So are the two exact verifiers of the newer parts — `graded_verify_stability.py` (§10.9) and `graded_verify_k4.py` (§11.9) — together with the instrument modules they declare and do not re-derive, and their stored logs. Its README carries the file manifest, naming the claim each file backs and the procedure for re-running each check. The failed design branches are retained there, because §6.5 is only checkable against them.

---

**Bibliography**

[1] G.-S. Cheon and S.-G. Hwang, *Maximization of a matrix function related to the Dittert conjecture*, Linear Algebra and its Applications **165** (1992), 153–165, `doi:10.1016/0024-3795(92)90234-2`.

[2] M. Putinar, *Positive polynomials on compact semi-algebraic sets*, Indiana University Mathematics Journal **42** (1993), 969–984.

[3] S.-G. Hwang, *On a conjecture of E. Dittert*, Linear Algebra and its Applications **95** (1987), 161–169. (Dittert's conjecture at $n = 3$.)

[4] R. Sinkhorn, *A problem related to the van der Waerden permanent theorem*, Linear and Multilinear Algebra **16** (1984), 167–173. (Dittert's conjecture at $n = 2$.)

[5] G.-S. Cheon and I. M. Wanless, *An update on Minc's survey of open problems involving permanents*, Linear Algebra and its Applications **403** (2005), 314–342.

[6] G.-S. Cheon and I. M. Wanless, *Some results towards the Dittert conjecture on permanents*, Linear Algebra and its Applications **436** (2012), 791–801, `doi:10.1016/j.laa.2010.08.041`. (Theorem 2.1 is Theorem E at $k = n$.)

[7] G.-S. Cheon and I. M. Wanless, *An interpretation of the Dittert conjecture in terms of semi-matchings*, Discrete Mathematics **307** (2007), `doi:10.1016/j.disc.2007.01.008`. (Not consulted; no result here depends on it.)

[8] Divya K. U. and K. Somasundaram, *Lih Wang's and Dittert's conjectures on permanents*, Special Matrices **12** (2024), 20240006, `doi:10.1515/spma-2024-0006`; cf. arXiv:2312.00464v1, whose abstract claims Dittert at $n = 4$. The published version makes no such claim.

[9] Z. Pang, *Proof of Dittert's conjecture for dimensions $n \geq 17$*, arXiv:2606.01531 (1 June 2026). Preprint, no DOI, not refereed.

[10] B. Kafidov, *Dittert's conjecture in dimension 16 via a joint-deficit scaling lemma*, arXiv:2607.19439 (21 July 2026). Preprint, no DOI, not refereed.

[11] Public repositories claiming Dittert cases, all created 21–25 July 2026, none refereed, all accessed 28 July 2026: `123ljh0bot/Dittert_Conjecture_in_Dimension_4` ($n = 4$, Lean 4 with an SOS certificate); `pedromnasc/dittert-conjecture-proof` ($n = 4, 5, 8$–$16$, and the `subdittert/` package quoted in §2.2); `lueluelue2006/dittert-conjecture-draft` and `lueluelue2006/dittert-n7-extension`, attributed to Hongyuan Lu ($n = 6$–$15$). Each is at `https://github.com/` followed by the owner and name shown.

[12] G.-S. Cheon and S. Yoon, *A note on the Dittert conjecture for permanents* (2006); G.-S. Cheon, *On the monotonicity of the Dittert function* (1993). The remaining two of the five papers citing [1]; neither works on the generalisation.

[13] J. B. Lasserre, *Global optimization with polynomials and the problem of moments*, SIAM Journal on Optimization **11** (2001), 796–817.

[14] K. Gatermann and P. A. Parrilo, *Symmetry groups, semidefinite programs, and sums of squares*, Journal of Pure and Applied Algebra **192** (2004), 95–128.

[15] E. Levin and V. Chandrasekaran, *Any-dimensional polynomial optimization via de Finetti theorems*, arXiv:2507.15632 (2025).

[16] The mathlib Community, *The Lean mathematical library*, CPP 2020.

[17] H. Tverberg, *On the permanent of a bistochastic matrix*, Mathematica Scandinavica **12** (1963), 25–35.

[18] S. Friedland, *A proof of a generalized van der Waerden conjecture on permanents*, Linear and Multilinear Algebra **11** (1982), 107–120. Zbl 0482.15003.

[19] P. McCullagh, *An asymptotic approximation for the permanent of a doubly stochastic matrix*, arXiv:1205.5723 (2012). The centred-core expansions of $\sigma_2, \sigma_3, \sigma_4$ in §10.3 are his, and the degree-five reduction specialises his general formula.

[20] G. P. Egorychev, *The solution of van der Waerden's problem for permanents*, Advances in Mathematics **42** (1981), 299–305.

[21] D. I. Falikman, *A proof of van der Waerden's conjecture on the permanent of a doubly stochastic matrix*, Matematicheskie Zametki **29** (1981), 931–938.